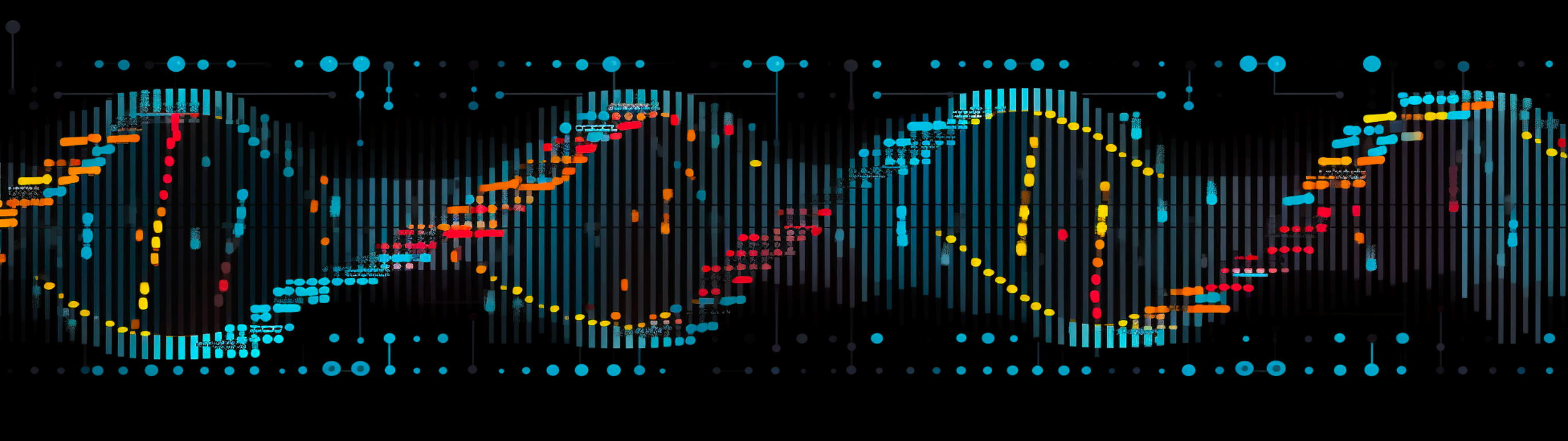




Introduction to bioinformatics

Chapter3 part2 Data and Resource

Jiaxing Chen

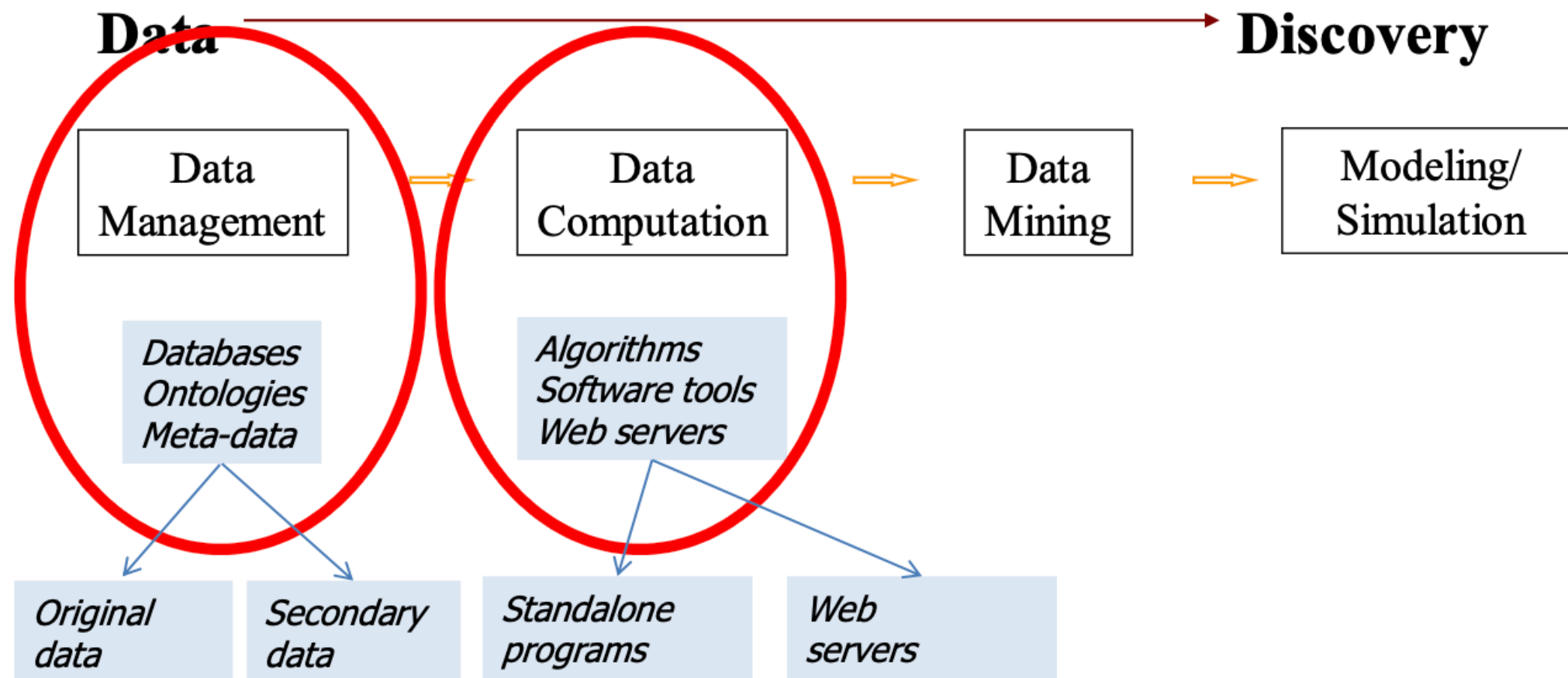
Outline

- **Biological file format**
- Resource
 - NCBI
 - EBI
 - UCSC
 - PDB
 - ...

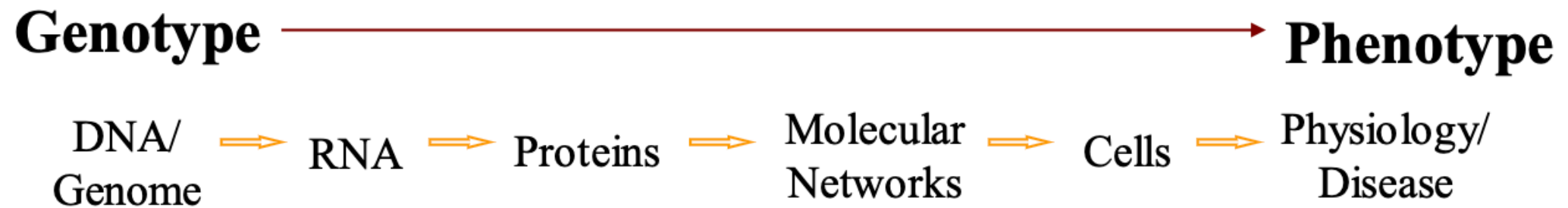
Definition of a biological database/resources

- The [NAR Database Issue](#) collects publications of established databases in the field
- Collection of data (and metadata) in the related format
 - structured
 - searchable (indexed)
 - updated periodically
 - entries mapped to unique identifiers, and cross-referenced
- Includes associated software necessary for DB access, update, search, visualisation (web)

The –informatics in Bioinformatics

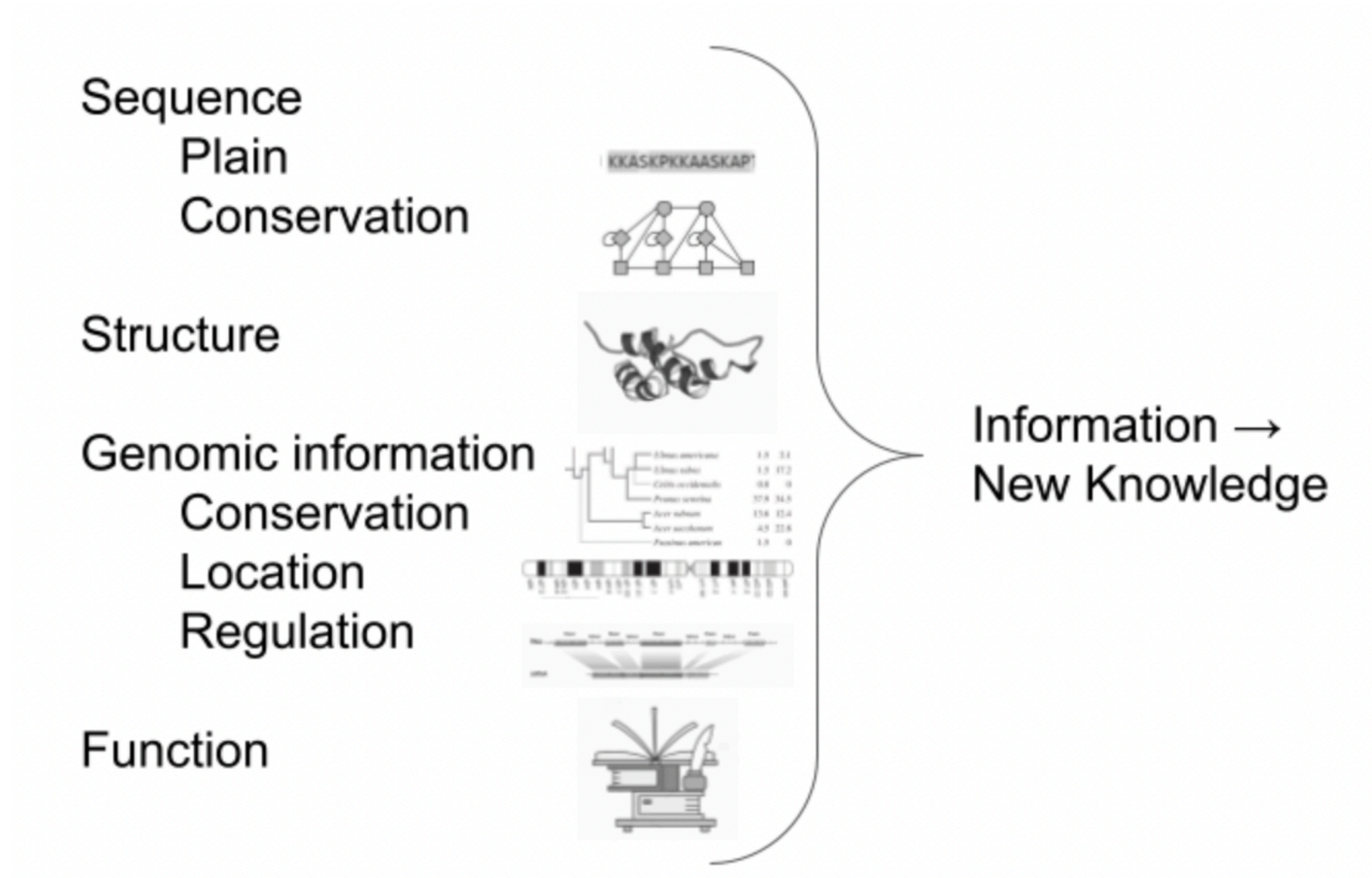


The Bio- in Bioinformatics



Biological knowledge

Understanding about biological entities comes from crossing the information from/to these different resources and formats



An intricate web of information exists around biological entities, and understanding them involves merging insights from various resources. A big part of some bioinformaticians' job is to integrate information from different databases and formats to gain a holistic understanding of biological entities.

Features of biological databases

- Data heterogeneity
- High volume of data
- Large scale data integration
- Data sharing / user visualisation and navigation
- Uncertainty / data quality measure needed
- Dynamic and subject to change

Possible classifications of biological databases

Data type

- Genome database
- Sequence database
- Structure database
- Pathway database
- Disease database
- ...

Data type

- Genome database
- Sequence database
- Structure database
- Pathway database
- Disease database
- ...

Data source

- Primary databases (GenBank, PDB)
- Secondary databases: analysed/aggregated results of the primary ones (UniProtKB)
- Composite database: non-redundant / filtered data (SwissProt)
- ...

Biological file formats

- Sequence formats
- Alignment formats
- Features/annotations formats
- Structure formats

Sequence formats

FASTA

File extensions: file.fa, file.fasta, file.fsa

Example:

```
>XR_002086427.1 Candida albicans SC5314 uncharacterized ncRNA (SCR1), ncRNA
```

```
TGGCTGTGATGGCTTTTAGCGGAAGCGCGCTGTTTCGCGTACCTGCTGTTTGTTGAAAATTTAAGAGCAAAGTGTCGGGCTCGATCCCTGCGAATTGAATTCTGAACGCTAGAGTAATCAGTGTCTTTCAAGTTC
```

Fasta format is a simple way of representing nucleotide or amino acid sequences of nucleic acids and proteins. This is a very basic format with two minimum lines. First line referred as comment line starts with ‘>’ and gives basic information about sequence. There is no set format for comment line. Any other line that starts with ‘;’ will be ignored. Lines with ‘;’ are not a common feature of fasta files. After comment line, sequence of nucleic acid or protein is included in standard one letter code. Any tabulators, spaces, asterisks etc in sequence will be ignored.

FASTQ

File extensions: ile.fastq, file.sanfastq, file.fq

Example:

[illegible]

Fastq format was developed by Sanger institute in order to group together sequence and its quality scores (Q: phred quality score). In fastq files each entry is associated with 4 lines.

- Line 1 begins with a '@' character and is a sequence identifier and an optional description.
- Line 2 Sequence in standard one letter code.
- Line 3 begins with a '+' character and is optionally followed by the same sequence identifier (and any additional description) again.
- Line 4 encodes the quality values for the sequence in Line 2, and must contain the same number of symbols as letters in the sequence.

Alignment formats

SAM (Sequence Alignment Map)

File extensions: file.sam

Example:

```
1:497:R:-272+13M17D24M 113 1 497 37 37M 15 100338662 0 CGGGTCTGACCTGAGGAGAACTGTGCTCCGCCTTCAG
19:20389:F:275+18M2D19M 99 1 17644 0 37M = 17919 314 TATGACTGCTAATAATACCTACACATGTTAGAACCAT >>>>>>
19:20389:F:275+18M2D19M 147 1 17919 0 18M2D19M = 17644 -314 GTAGTACCAACTGTAAGTCCTTATCTTCATACTTTGT
9:21597+10M2I25M:R:-209 83 1 21678 0 8M2I27M = 21469 -244 CACCACATCACATATACCAAGCCTGGCTGTGTCTTCT <;9<<5>
```

The SAM Format is a text format for storing sequence data in a series of tab delimited ASCII columns. Most often it is generated as a human readable version of its sister BAM format, which stores the same data in a compressed, indexed, binary form.

SAM format files are generated following mapping of the reads to reference sequence. It is TAB-delimited text format with header and a body. Header lines start with '@' while alignment lines do not. Header hold generic information on SAM file along with version information, if the file is sorted, information on reference sequence, etc. The alignment records constitute the body of the file. Each alignment line/record has 11 mandatory fields describing essential alignment information.

Alignment formats

BAM (Binary Alignment/Map)

File extensions: file.bam

A BAM file is the compressed binary version of the Sequence Alignment/Map (SAM).

a compact and indexable representation of nucleotide sequence alignments. The data between SAM and BAM is exactly same. Being Binary BAM files are small in size and ideal to store alignment files. Require samtools to view the file.

Features/annotations formats

VCF (Variant Calling Format/File)

File extensions: file.vcf

Example:

```
##fileformat=VCFv4.2
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
...
20 14370 rs6054257 G A 29 PASS NS=3;DP=14;AF=0.5;DB;H2 GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20 17330 . T A 3 q10 NS=3;DP=11;AF=0.017 GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3 0/0:41:3
20 1110696 rs6040355 A G,T 67 PASS NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2 2/2:35:4
20 1230237 . T . 47 PASS NS=3;DP=13;AA=T GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20 1234567 microsat1 GTC G,GTCT 50 PASS NS=3;DP=9;AA=G GT:GQ:DP 0/1:35:4 0/2:17:2 1/1:40:3
```

VCF is a text file format with a header (information VCF version, sample etc) and data lines constitute the body of file.

Features/annotations formats

GFF (General Feature Format or Gene Finding Format)

File extensions: file.gff2, file.gff3, file.gff

Example (GFF2):

```
browser position chr22:10000000-10025000

browser hide all

track name=regulatory description="TeleGene(tm) Regulatory Regions"

visibility=2

chr22 TeleGene enhancer 10000000 10001000 500 + . touch1

chr22 TeleGene promoter 10010000 10010100 900 + . touch1

chr22 TeleGene promoter 10020000 10025000 800 - . touch2
```

GFF (General Feature Format or Gene Finding Format). GFF can be used for any kind of feature (Transcripts, exon, intron, promoter, 3' UTR, repetitive elements etc) associated with the sequence, whereas GTF is primarily for genes/transcripts. GFF3 is the latest version and an improvement over GFF2 format. However, many databases are still not equipped to handle GFF3 version. The differences will be explained later in text.

The GFF format has 9 mandatory columns and they are TAB separated.

- Col. 1 Reference Sequence
- Col. 2 Source
- Col. 3 Feature
- Col. 4 Start
- Col. 5 End
- Col. 6 Score
- Col. 7 Strand
- Col. 8 Frame (GFF2 and GTF) or Phase (GFF3)
- Col. 9 Attribute or Group field

Features/annotations formats

BED (Browser Extensible Data)

The BED (Browser Extensible Data) file format includes information about sequences that can be visualized in a genome browser; a feature called an annotation track. BED files are tabs-delimited and include 12 fields (columns) of data.

Example of fields: name of chromosome or scaffold, starting position in the chromosome, the ending position...

PSI-MI

The PSI MI format is a data exchange format for molecular interactions.

Example of fields: interaction detection method, biological role, experimental features, location of the interaction, ...

PED

File extensions: file.ped

PED is a file format for pedigree analysis, which creates a familial relationship between different samples.

Structure formats

PDB (Protein Data Bank formats)

File extensions: file.pdb

PDB file formats contain atomic coordinates and are used for storing 3D protein structures by the Protein Data Bank.

Example:

```
COMPND      UNNAMED
AUTHOR      GENERATED BY OPEN BABEL 2.3.2
ATOM        1  N   ALA A   1           0.000  0.000  0.000  1.00  0.00           N
ATOM        2  CA  ALA A   1           1.456  0.000  0.000  1.00  0.00           C
ATOM        3  C   ALA A   1           1.930  0.000  1.463  1.00  0.00           C
ATOM        4  O   ALA A   1           1.160  0.000  2.421  1.00  0.00           O
...
CONNECT     101    98
CONNECT     102    94   103
CONNECT     103   102
MASTER
            0    0    0    0    0    0    0    0    0   103    0   103    0
END
```

Other formats

CSV

CSV (.csv file format) files stands for comma separated value and is a text file, where each line is a row and columns are delimited with a comma. It can store different types of sequencing data and can be opened using common spreadsheet programs.

JSON

JSON (JavaScript Object Notation) is a common file format for many other industries, but is used in a growing number of bioinformatics applications and web resources.

And the list of generic file formats goes on...

Why Are There So Many Different Types?

The many different ways of generating and using biological data have given rise to the diversity previously described. These file formats have their own specific use cases depending on:

- Compatibility with specific software
- Data processing, parsing, and human readability needs
- Efficiency for storage

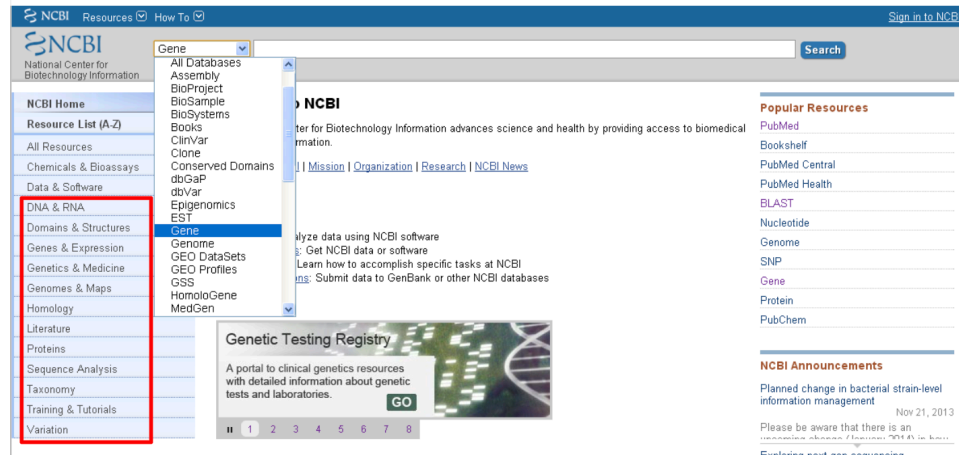
In conclusion, the multitude of biological file formats arises from the diverse needs and characteristics of biological data.

Key Points

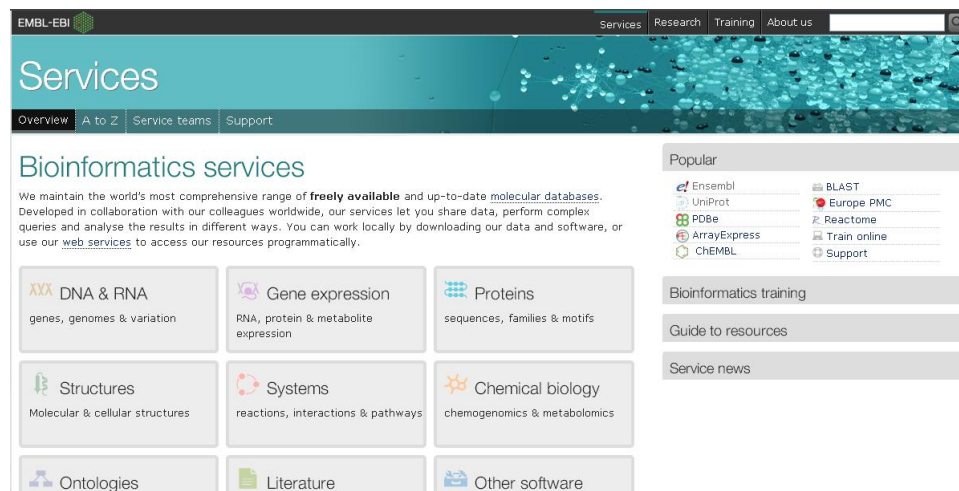
- Biological data is multi-layered. E.g. the information about one gene can actually regard multiple different biological entities: the variability of its sequence, the derived protein, the diseases associated etc.
- Consequently, several different sources of information can be identified and used to describe a biological entity, as well as several different file formats.

Centralized vs. individual resources

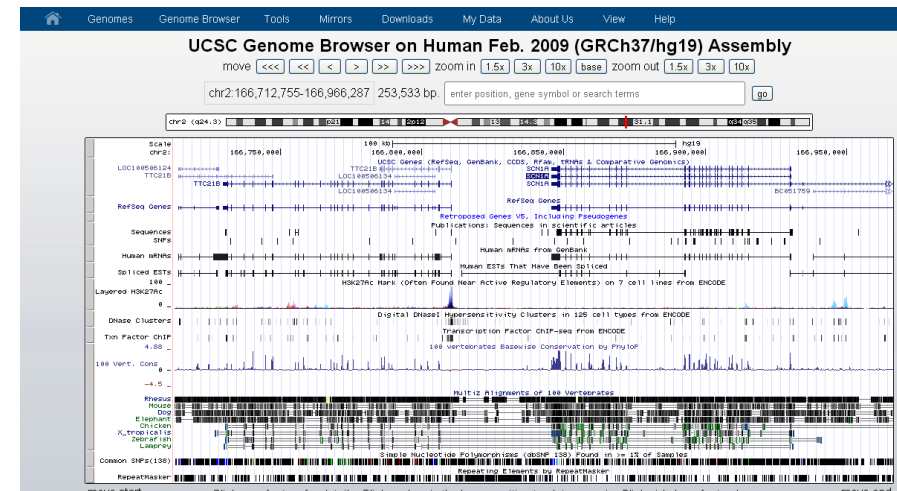
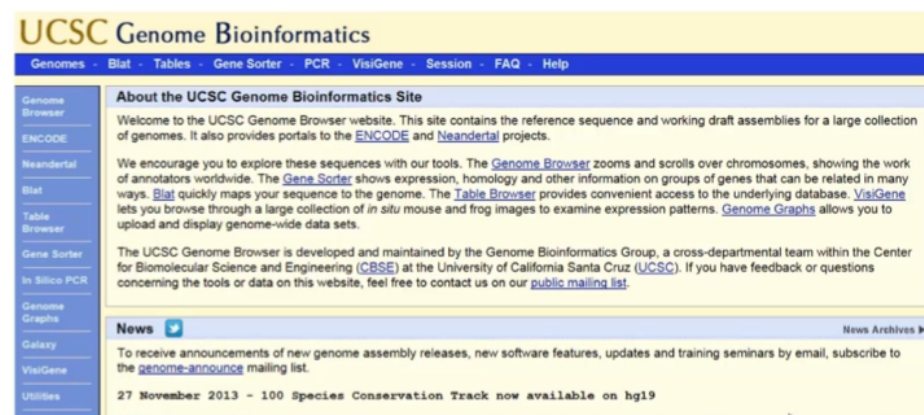
NCBI: <http://www.ncbi.nlm.nih.gov/>



EBI: <http://www.ebi.ac.uk/services>



UCSC Genome Bioinformatics (<http://genome.ucsc.edu/>)



NCBI/EBI/UCSC Summary Table

	NCBI	EBI	UCSC
Tools	BLAST	BLAST Exonerate ClustalW2	BLAT In-Silico PCR
Data Repository	GenBank GEO SRA	ArrayExpress ENA PDBe	ENCODE
DNA/Genome	Genome	Ensembl	Ideogram Recombination Rate GC Content

NCBI/EBI/UCSC Summary Table (Cont'd)

	NCBI	EBI	UCSC
DNA/Gene	Gene	Ensembl	UCSC Genes GENCODE RefSeq Genes
RNA	RefSeq	Ensembl	mRNAs ESTs UniGene
Proteins	Protein, RefSeq Conserved Domain	UniProt, InterPro, PRIDE (mass spec)	

NCBI/EBI/UCSC Summary Table (Cont'd)

	NCBI	EBI	UCSC
Expression	UniGene	Expression Atlas	Affy Exon Array Caltech RNA-seq Allen Brain
Regulation			Transcription TFBS Epigenetics DNaseI HS
Literature	PubMed		
Ontology		Gene Ontology	

NCBI/EBI/UCSC Summary Table (Cont'd)

	NCBI	EBI	UCSC
Comparative Genomics	Taxonomy HomoloGene	*Ensembl	Conservation Neandertal
Variation	dbSNP dbVar	*Ensembl	SNPs DGV RepeatMasker
Disease	OMIM MeSH dbGaP ClinVar		GAD COSMIC ClinVar GWAS Catalog QTLs

NCBI access

- AllResources – Database:

http://www.ncbi.nlm.nih.gov/guide/all/#databases_

- Tools:

http://www.ncbi.nlm.nih.gov/guide/all/#tools_

- Resource List (A-Z)

- <http://www.ncbi.nlm.nih.gov/guide/sitemap/>

EBI access

- Bioinformatics services

- by topic:

<http://www.ebi.ac.uk/services>

- DNA&RNA, Geneexpression, Proteins, Structures, Systems, Chemical biology, Ontologies, Literature, ...

- Popular: Ensembl, UniProt, PDBe, ArrayExpress, ChEMBL, BLAST, Europe PMC, Reactome, Train online, ...

- by name(A-Z):

<http://www.ebi.ac.uk/services/all>

UCSC access

- Table Browser

- <http://genome.ucsc.edu/cgi-bin/hgTables>

- GenomeBrowser

- <http://genome.ucsc.edu/cgi-bin/hgTracks> • trackhubs

- configure

Examples of individual Resources

Analyses		Databases and tools
DNA/Genome	Gene prediction	GENSCAN, Glimmer
	Genetic and Somatic Variations	HGMD, COSMIC, SIFT, PolyPhen, SAPRED
Expression regulation	Transcription factors & binding	TRANSFAC, PlantTFDB
	RNA annotation	Rfam
	microRNA annotation	miRBase
Epigenetics	DNA methylation	MethylomeDB
Pathway & network	Pathway	KEGG, PANTHER, BioCyc, REACTOME, KOBAS, DAVID
	Interaction network	PID, STRING
Evolution	Conservation	GERP++, PHYML

Examples of individual resources for mass spectrometry proteomics analyses

	Databases and tools
Mass spec data repository	The Global Proteome Machine
Peptide identification by database search	Sequest, Mascot, ProteinProspector, pFind, PEAKS, Byonic, Proteome Discoverer, SpectrumMill, Masslynx, X!Tandem
<i>de novo</i> peptide identification	pNovo, PEAKS, PepNovo
Quantitation	MaxQuant, Census

Examples of Individual Resources for Protein 3D Structures

Analyses	Databases and Tools
Protein 3D structures	PDB
Homology Modeling	Modeller, Swiss-Model, I-TASSER
Fold Recognition	3D-PSSM, Phyre2, PROSPECT
<i>ab initio</i> folding	QUARK, Rosetta
Protein 3D models	Swiss-Model Repository
Nucleotide structure	Mfold, PDB
Nucleotide interaction	RNAhybrid

Examples of Individual Resources for NGS analysis

Analyses		Databases and tools
Read mapping	Reads mapping (DNA)	BWA, Bowtie/Bowtie2
	Reads mapping (RNA)	TopHat/TopHat2
	Reads mapping related utilities	GATK, FastQC, RNA-SeQC, SAMtools, Picard
Assembly	de novo genomic assembly	Velvet, SOAP de novo
	de novo transcriptome assembly	Trinity, Velvet + Oases
	reference-based transcriptome assembly	TopHat + Cufflinks
Visualization	Genome Browser	GBrowse, JBrowse, IGV
Variant calling	SNP	GATK, SOAPsnp, SAMtools
	Indel	Pindel
	CNV and structural variation	CNVnator, BIC-seq, SVMerge, mrCaNaVaR, ExomeCNV, CoNIFER, HMMcopy, Control-FREEC
Expression	Differential expression	Cuffdiff, DESeq/DESeq2
ChIP-Seq	Peak calling	MACS

Other examples of individual resources & utilities

Purpose		Databases and tools
Model Organisms	Genome and gene annotations	Flybase, Wormbase, ZFIN, TAIR
Large-scale studies	Cancer	TCGA, CGP
	Epigenetics	Roadmap Epigenomics Project
	Brain	Allen Brain Atlas, Human Connectome project
Tools to assist wet-lab experiments	Primer design	Primer3/Primer3Plus, Electronic PCR
Software programming utilities	Sequence analysis tool package	EMBOSS
	R package	Bioconductor
	Perl package	BioPerl
	Python package	BioPython
Workflow	Workflow platform	Galaxy
	Workflow construction	Taverna

NAR database issue & web server issue

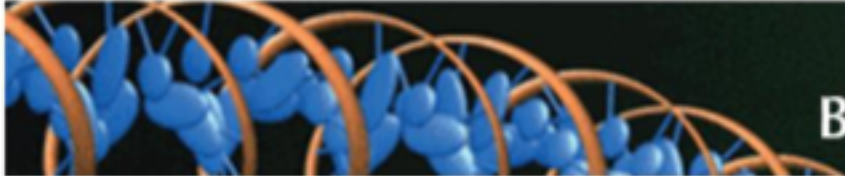
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Nucleic Acids Research

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Oxford Journals > Life Sciences > Nucleic Acids Research > Volume 41, Issue D1



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Volume 41 Issue D1 January 2013

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January 2013 41 (D1)

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Current Issue

November 2013 41 (21)

NCBI: <http://www.ncbi.nlm.nih.gov/>

The screenshot shows the NCBI homepage with the 'Gene' dropdown menu open. The menu lists various databases and resources, with 'Gene' highlighted. The main content area features a 'Genetic Testing Registry' banner and a 'Popular Resources' sidebar.

NCBI Resources ▾ How To ▾ Sign in to NCBI

NCBI
National Center for Biotechnology Information

NCBI Home
Resource List (A-Z)

- All Resources
- Chemicals & Bioassays
- Data & Software
- DNA & RNA
- Domains & Structures
- Genes & Expression
- Genetics & Medicine
- Genomes & Maps
- Homology
- Literature
- Proteins
- Sequence Analysis
- Taxonomy
- Training & Tutorials
- Variation

Gene ▾

- All Databases
- Assembly
- BioProject
- BioSample
- BioSystems
- Books
- ClinVar
- Clone
- Conserved Domains
- dbGaP
- dbVar
- Epigenomics
- EST
- Gene**
- Genome
- GEO DataSets
- GEO Profiles
- GSS
- HomoloGene
- MedGen

NCBI
The National Center for Biotechnology Information advances science and health by providing access to biomedical information.
[Home](#) | [Mission](#) | [Organization](#) | [Research](#) | [NCBI News](#)

Popular Resources

- [PubMed](#)
- [Bookshelf](#)
- [PubMed Central](#)
- [PubMed Health](#)
- [BLAST](#)
- [Nucleotide](#)
- [Genome](#)
- [SNP](#)
- [Gene](#)
- [Protein](#)
- [PubChem](#)

NCBI Announcements

Planned change in bacterial strain-level information management
Nov 21, 2013

Please be aware that there is an upcoming change (January 2014) in how we handle bacterial strain-level information. For more information, see the announcement on the NCBI website.

Exploring next-gen sequencing

Genetic Testing Registry
A portal to clinical genetics resources with detailed information about genetic tests and laboratories. [GO](#)

1 2 3 4 5 6 7 8

Examples of resources at NCBI

Data Repository	GenBank, dbEST, GEO, SRA
DNA	Genome, Gene
Comparative Genomics	Taxonomy, HomoloGene
Genetic Variation	dbSNP, dbVar
Disease Mutations	OMIM, dbGaP, ClinVar
RNA	RefSeq, UniGene
Proteins	Protein, RefSeq, Conserved Domain
Literature	PubMed, MeSH
Tools	BLAST

NCBI-Genome

<http://www.ncbi.nlm.nih.gov/genome/>

 NCBI [Resources](#)  [How To](#)  [Sign in to NCBI](#)

Genome [Limits](#) [Advanced](#) [Help](#)



Genome

This resource organizes information on genomes including sequences, maps, chromosomes, assemblies, and annotations.

Using Genome

- [Help](#)
- [Browse by Organism](#)
- [Download / FTP](#)
- [Submit a genome](#)

Custom resources

- [Human Genome](#)
- [Microbes](#)
- [Organelles](#)
- [Plants](#)
- [Viruses](#)

Other Resources

- [Assembly](#)
- [BioProject](#)
- [BioSample](#)
- [Map Viewer](#)
- [Protein Clusters](#)

Genome Tools

- [BLAST the Human Genome](#)
- [Genomic groups BLAST](#)
- [NCBI remap](#)
- [Genome Decoration Page](#)

Genome Annotation and Analysis

- [Eukaryotic Genome Annotation](#)
- [Prokaryotic Genome Annotation](#)
- [PASC \(Pairwise Sequence Comparison\)](#)
- [TaxPlot \(3-way Genome Comparison\)](#)

External Resources

- [GOLD - Genomes Online Database](#)
- [Ensembl Genome Browser](#)
- [Bacteria Genomes at Sanger](#)
- [Large-Scale Genome Sequencing \(NHGRI\)](#)

http://www.ncbi.nlm.nih.gov/mapview/maps.cgi?ORG=hum&M=

NCBI Map Viewer

Human genome overview page (Annotation Release 105)
Human genome overview page (Build 36.3)
Map Viewer Home
Map Viewer Help
Human Maps Help
FTP
Data As Table View
Maps & Options
Region Shown
out zoom in
You are here:
Ideogram

Search Find Find in this View Advanced Search

Homo sapiens (human) Annotation Release 105 (Current) [BLAST human sequences](#)

Chromosome: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 X Y MT

Master Map: Genes On Sequence [Summary of Maps](#) [Maps & Options](#)

Region Displayed: 0 191M bp

☒ Ideogram ☒ Hs ☒ UniG ☒ Genes_seq

Symbol	Q	Links	E	Cyto	Description
FGFR3	+	OMIM HGNC sv pr dl ev hm sts CCDS SNP	best RefSeq	4p16.3	fibroblast growth factor receptor 3
HTT	+	OMIM HGNC sv pr dl ev hm sts CCDS SNP	best RefSeq	4p16.3	huntingtin
PROM1	+	OMIM HGNC sv pr dl ev hm sts CCDS SNP	best RefSeq	4p15.32	prominin 1
PPARGC1A	+	OMIM HGNC sv pr dl ev hm sts CCDS SNP	best RefSeq	4p15.1	peroxisome proliferator-activated r
PDGFRA	+	OMIM HGNC sv pr dl ev hm sts CCDS SNP	best RefSeq	4q12	platelet-derived growth factor recej
KIT	+	OMIM HGNC sv pr dl ev hm sts CCDS SNP	best RefSeq	4q12	v-kit Hardy-Zuckerman 4 feline sar
KDR	+	OMIM HGNC sv pr dl ev hm sts CCDS SNP	best RefSeq	4q11-q12	kinase insert domain receptor (a ty)
ALB	+	OMIM HGNC sv pr dl ev hm sts CCDS SNP	best RefSeq	4q13.3	albumin
IL8	+	OMIM HGNC sv pr dl ev hm sts CCDS SNP	best RefSeq	4q13-q21	interleukin 8
CXCL10	+	OMIM HGNC sv pr dl ev hm sts CCDS SNP	best RefSeq	4q21	chemokine (C-X-C motif) ligand 10
SPP1	+	OMIM HGNC sv pr dl ev hm sts CCDS SNP	best RefSeq	4q22.1	secreted phosphoprotein 1
ABCF2	+	OMIM HGNC sv pr dl ev hm sts CCDS SNP	best RefSeq	4q22	ATP-binding cassette, sub-family C

NCBI-Nucleotide/Protein (RefSeq)

www.ncbi.nlm.nih.gov/nuccore/ www.ncbi.nlm.nih.gov/protein/

[Sign in to NCBI](#)

Protein

Protein



Search

[Advanced](#)

[Help](#)



Protein

The Protein database is a collection of sequences from several sources, including translations from annotated coding regions in GenBank, RefSeq and TPA, as well as records from SwissProt, PIR, PRF, and PDB. Protein sequences are the fundamental determinants of biological structure and function.

Using Protein

[Quick Start Guide](#)

[FAQ](#)

[Help](#)

[GenBank FTP](#)

[RefSeq FTP](#)

Protein Tools

[BLAST](#)

[LinkOut](#)

[E-Utilities](#)

[Blink](#)

[Batch Entrez](#)

Other Resources

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[RefSeq Home](#)

[CDD](#)

[Structure](#)

You are here: [NCBI](#) > [Proteins](#) > Protein Database

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[Genetic Testing Registry](#)

[PubMed Health](#)

NCBI INFORMATION

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NCBI-Nucleotide/Protein (RefSeq): [NM_*, NP_*]

www.ncbi.nlm.nih.gov/nuccore/ www.ncbi.nlm.nih.gov/protein/ [Change Region](#)

transcript variant 1, mRNA

NCBI Reference Sequence: NM_001017424.2

[FASTA](#) [Graphics](#)

[Go to:](#) ☐

LOCUS NM_001017424 3597 bp mRNA linear PRI 17-NOV-2013
DEFINITION Homo sapiens potassium channel, subfamily K, member 2 (KCNK2),
transcript variant 1, mRNA.
ACCESSION NM_001017424
VERSION NM_001017424.2 GI:126365744
KEYWORDS RefSeq.
SOURCE Homo sapiens (human)
ORGANISM [Homo sapiens](#)
Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini;
Catarrhini; Hominidae; Homo.
REFERENCE 1 (bases 1 to 3597)

[Customize view](#)

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[Find in this Seq](#)

Articles about gene

[Regulation of int
n J Physiol Lung](#)

NCBI-Gene

<http://www.ncbi.nlm.nih.gov/gene/>

Display Settings: ☒ Full Report

Send to: ☐

ABCA1 ATP-binding cassette, sub-family A (ABC1), member 1 [*Homo sapiens* (human)]

Gene ID: 19, updated on 21-Nov-2013

Summary

Official Symbol ABCA1 provided by [HGNC](#)
Official Full Name ATP-binding cassette, sub-family A (ABC1), member 1 provided by [HGNC](#)
Primary source [HGNC:29](#)
See related [Ensembl:ENSG00000165029](#); [HPRD:02501](#); [MIM:600046](#); [Vega:OTTHUMG00000020417](#)
Gene type protein coding
RefSeq status REVIEWED
Organism [Homo sapiens](#)
Lineage Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo
Also known as TGD; ABC1; CERP; ABC-1; HDLDT1
Summary The membrane-associated protein encoded by this gene is a member of the superfamily of ATP-binding cassette (ABC) transporters. ABC proteins transport various molecules across extra- and intracellular membranes. ABC genes are divided into seven distinct subfamilies (ABC1, MDR/TAP, MRP, ALD, OABP, GCN20, White). This protein is a member of the ABC1 subfamily. Members of the ABC1 subfamily comprise the only major ABC subfamily found exclusively in multicellular eukaryotes. With cholesterol as its substrate, this protein functions as a cholesterol efflux pump in the cellular lipid removal pathway. Mutations in this gene have been associated with Tangier's disease and familial high-density lipoprotein deficiency. [provided by RefSeq, Jul 2008]

Genomic context

Location: 9q31.1
Sequence: Chromosome: 9; NC_000009.11 (107543283..107690527, complement)

See ABCA1 in [Epigenomics](#), [MapView](#)

Chromosome 9 - NC_000009.11

107509969 ▶

▶ 108200785

Table of contents

[Summary](#)
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Human genes: GeneCards <http://www.genecards.org/>

The screenshot displays the GeneCards website, which is a comprehensive database of human genes. The header features the GeneCards logo, the Weizmann Institute of Science logo, and the LifeMap Sciences logo. A navigation bar includes links to Home, GeneCards Guide, Suite, Terms and Conditions, About Us, User Feedback, and Mirror sites. A search bar allows users to search by keyword(s) or use advanced search options. The main content area is divided into several sections: Affiliated Sites (MalaCards, GeneCards Suite, GeneDecks, GeneLaCart, GeneLoc), Explore GeneCards (About GeneCards, Extract information for many genes at once, View Sample Gene, View Random Gene), and News and Views (Our Trendy Genes, New In V3.11, In our pipeline, Collaborations, Commentary, Versions). The 'View Sample Gene' section shows details for TNFRSF10B (tumor necrosis factor receptor superfamily, member 10b) and MAPK7 (mitogen activated protein kinase 7). A grid of buttons provides access to various data sections for each gene, including Aliases, Drugs, Genome view, Pathways, Publications, Databases, Expression, Interactions, Paralogs, Summaries, Disorders, External search, IP/Proteins, Products, Transcripts, Domains, Function, Orthologs, Proteins, and Variants.

GeneCards
The Human Gene Compendium

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Affiliated Sites

- MalaCards: The Human Molecule Compendium
- GeneCards Suite
- GeneDecks: Personalized Gene Suite
- GeneLaCart: BETA Batch Queries
- GeneLoc: Gene Coordinates

Explore GeneCards

About GeneCards®: GeneCards is a searchable, integrated database of human genes that provides comprehensive, updated, and user-friendly information on all known and predicted [more...](#)

Extract information for many genes at once: [GeneLaCart](#) [GeneDecks](#) [Hot genes](#) [Disease genes](#)

View Sample Gene

GeneCards Sections:

Aliases	Drugs	Genome view	Pathways	Publications
Databases	Expression	Interactions	Paralogs	Summaries
Disorders	External search	IP/Proteins	Products	Transcripts
Domains	Function	Orthologs	Proteins	Variants

View Random Gene

Category: Protein-coding

GIFTS Group: High

TNFRSF10B
tumor necrosis factor receptor superfamily, member 10b

MAPK7
(GIFTS: 73)
mitogen activated protein kinase 7

News and Views

Our Trendy Genes

published in [GEN](#)

New In V3.11:

- [IL13R1A1 ligands](#)
- [MAXOR protein expr](#)
- [Gene/protein families](#)
- [more...](#)

In our pipeline:

- [PathCards](#)

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- [RNAcentral](#)

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Versions:

- Current: 3.11
- 17 Nov 2013
- Previous: 3.10

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SRA

The Sequence Read Archive (SRA) stores raw sequencing data from the next generation of sequencing platforms including Roche 454 GS System®, Illumina Genome Analyzer®, Applied Biosystems SOLiD® System, Helicos Heliscope®, Complete Genomics®, and Pacific Biosciences SMRT®.

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[Trace Assembly](#)

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<http://www.ncbi.nlm.nih.gov/sra>

ERX033410: Foxa1 and Foxa2 are essential for gender dimorphism in liver cancer

1 ILLUMINA (Illumina Genome Analyzer Iix) run: 40.9M spots, 1.5G

[Change access](#)

Accession: ERX033410

Experiment design: Foxa1 and Foxa2 are essential for gender dim

Submission: ERA070841 by UNIVERSITY OF PENNSYLVANIA

Study summary: Foxa1 and Foxa2 are essential for gender dimorpl

[Study](#) • [All experiments \(more...\)](#)

Sample: Treated-Male-Control ([ERS074997](#)) ([less...](#))

Organism: [Mus musculus](#)

Attributes:

OrganismPart: liver

Genotype: Foxa1loxP/loxP;Foxa2loxP/loxP

StrainOrLine: 129J/BL6

Organism: Mus musculus

Sex: male

Library: MDAR ([more...](#))

Platform: Illumina ([less...](#))

Instrument model: Illumina Genome Analyzer Iix

Spot descriptor:

1 forward

Experiment attributes:

Experimental Factor: ORGANISM: Mus musculus

Experimental Factor: SEX: male

Experimental Factor: TREATMENT: Treated

Experimental Factor: GENOTYPE: Foxa1loxP/loxP;Foxa2loxP/loxP

Experimental Factor: IMMUNOPRECIPITATE: AR

Total: 1 run, 40.9M spots, 1.5G bases, [918.8Mb](#)  

#	Run	# of Spots	# of Bases	Size
1.	ERR056365	40,948,224	1.5G	918.8Mb

ID: 197484

Run ERR056365 (Foxa1 and Foxa2 are essential for gender dimorphism in liver cancer)

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Run	Spots	Bases	Size	GC content	Published	Access Type
ERR056365	40.9 M	1.5 Gbp	963.4 M	43.8%	2012-06-26	public

Quality graph ([bigger](#))

This run has 1 read per spot:

L=36, 100%

[Legend](#)

Experiment	Library					
ERX033410	Name	Platform	Strategy	Source	Selection	Layout
	MDAR	Illumina	ChIP-Seq	GENOMIC	ChIP	SINGLE

[Show design](#)

Biosample	Sample name	Title
ERS074997	E-MTAB-805:Treated-Male-Control	Treated-Male-Control

Bioproject	SRA Study	Title
	ERP001029	Foxa1 and Foxa2 are essential for gender dimorphism in liver cancer

[Metadata](#) [Reads](#) [Download](#)

Object		.sra		
Run	ERR056365	963.4 Mb	HTTP	FTP
Experiment	ERX033410	963.4 Mb	HTTP	FTP
Study	ERP001029	12.0 Gb	HTTP	FTP

(use [Aspera plugin](#) for fast download)

NCBI-Taxonomy

<http://www.ncbi.nlm.nih.gov/taxonomy/>

NCBI Taxonomy Browser

Entrez PubMed Nucleotide Protein Genome Structure PMC Taxonomy

Search for as complete name ☐ lock

Display levels using filter:

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Systematic names

Lineage (full): [root](#); [cellular organisms](#); [Eukaryota](#); [Opisthokonta](#); [Metazoa](#); [Eumetazoa](#); [Bilateria](#); [Deuterostomia](#); [Chordata](#); [Craniata](#); [Vertebrata](#); [Gnathostomata](#); [Teleostomi](#); [Euteleostomi](#); [Sarcopterygii](#); [Dipnotetrapodomorpha](#); [Tetrapoda](#); [Amniota](#); [Mammalia](#); [Theria](#); [Eutheria](#); [Euarchontoglires](#); [Primates](#); [Haplorrhini](#); [Simiiformes](#); [Catarrhini](#); [Hominoidea](#); [Hominidae](#); [Homininae](#); [Homo](#)

◦ [Homo sapiens](#) (human) Click on organism name to get more information.

- [Homo sapiens neanderthalensis](#) (Neandertal)
- [Homo sapiens ssp. Denisova](#) (Denisova hominin)

Disclaimer: The NCBI taxonomy database is not an authoritative source for nomenclature or classification; please consult the relevant scientific literature for the most reliable information.

Comments and questions to info@ncbi.nlm.nih.gov

Homo sapiens

Taxonomy ID: 9606
Genbank common name: **human**
Inherited blast name: **primates**
Rank: species
Genetic code: [Translation table 1 \(Standard\)](#)
Mitochondrial genetic code: [Translation table 2 \(Vertebrate Mitochondrial\)](#)
Other names:
common name: **man**
authority: **Homo sapiens Linnaeus, 1758**
Lineage (full)
[cellular organisms](#); [Eukaryota](#); [Opisthokonta](#); [Metazoa](#); [Eumetazoa](#); [Bilateria](#); [Deuterostomia](#); [Chordata](#); [Craniata](#); [Vertebrata](#); [Gnathostomata](#); [Teleostomi](#); [Euteleostomi](#); [Sarcopterygii](#); [Dipnotetrapodomorpha](#); [Tetrapoda](#); [Amniota](#); [Mammalia](#); [Theria](#); [Eutheria](#); [Euarchontoglires](#); [Primates](#); [Haplorrhini](#); [Simiiformes](#); [Catarrhini](#); [Hominoidea](#); [Hominidae](#); [Homininae](#); [Homo](#)

This is the top level of the taxonomy database maintained by NCBI/GenBank. You can explore any of the taxa listed below by clicking it.

- [Archaea](#)
- [Bacteria](#)
- [Eukaryota](#)
- [Viroids](#)
- [Viruses](#)
- [Other](#)
- [Unclassified](#)

These are direct links to some of the organisms commonly used in molecular research projects:

Arabidopsis thaliana	Escherichia coli	Pneumocystis carinii
Bos taurus	Hepatitis C virus	Rattus norvegicus
Caenorhabditis elegans	Homo sapiens	Saccharomyces cerevisiae
Chlamydomonas reinhardtii	Mus musculus	Schizosaccharomyces pombe
Danio rerio (zebrafish)	Mycoplasma pneumoniae	Takifugu rubripes
Dictyostelium discoideum	Oryza sativa	Xenopus laevis

Entrez records		
Database name	Subtree links	Direct links
Nucleotide	10,401,809	10,401,779
Nucleotide EST	8,704,844	8,704,844
Nucleotide GSS	1,729,196	1,727,870
Protein	800,192	800,044
Structure	23,839	23,839
Genome	1	1
Popset	15,254	15,254
SNP	73,357,341	73,357,341
Domains	11	11
GEO Datasets	598,345	598,345
UniGene	130,056	130,056
UniSTS	328,873	328,873
PubMed Central	15,467	15,448
Gene	199,013	198,940
HomoloGene	18,473	18,473
SRA Experiments	114,540	114,510
Probe	27,841,436	27,841,435

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NCBI-Taxonomy Statistics

<http://www.ncbi.nlm.nih.gov/taxonomy/>

Taxonomy Nodes (all dates)

Ranks:	higher taxa	genus	species	lower taxa	total
Archaea	703	291	970	0	1,964
Bacteria	6,769	5,257	26,278	964	39,268
Eukaryota	69,307	100,515	539,227	38,309	747,358
Fungi	6,335	7,688	59,812	1,584	75,419
Metazoa	50,074	71,780	280,733	19,250	421,837
Viridiplantae	8,685	17,003	183,434	17,080	226,202
Viruses	2,258	2,794	5,976	793	11,821
All taxa	79,068	108,858	572,437	40,066	800,429

Dates: [2015](#) [2016](#) [2017](#) [2018](#) [2019](#) [2020](#) [2021](#) [2022](#) [2023](#) [2024](#) [all dates](#)

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Nucleic Acids Res. 2011 Jul;39(Web Server Issue):W316-22. doi: 10.1093/nar/gkr493.

KOBAS 2.0: a web server for annotation and identification of enriched pathways and diseases.

Xie G, Mao X, Huang J, Ding Y, Wu J, Dong S, Kang L, Gao G, Li GY, Wei L

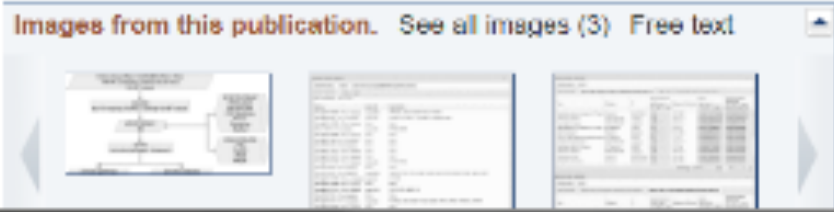
Center for Bioinformatics, State Key Laboratory of Protein and Plant Gene Research, College of Life Sciences, Peking University, Beijing, China

Abstract

High-throughput experimental technologies often identify dozens to hundreds of genes related to, or changed in, a biological or pathological process. From these genes one wants to identify biological pathways that may be involved and diseases that may be implicated. Here, we report a web server, KOBAS 2.0, which annotates an input set of genes with putative pathways and disease relationships based on mapping to genes with known annotations. It allows for both ID mapping and cross-species sequence similarity mapping. It then performs statistical tests to identify statistically significantly enriched pathways and diseases. KOBAS 2.0 incorporates knowledge across 1327 species from 5 pathway databases (KEGG PATHWAY, PID, BioCyc, Reactome and Panther) and 5 human disease databases (OMIM, KEGG DISEASE, FunDO, GAD and NHGRI GWAS Catalog). KOBAS 2.0 can be accessed at <http://kobas.cbi.pku.edu.cn>.

PMID: 21715386 [PubMed - indexed for MEDLINE] PMCID: PMC3125809 **Free PMC Article**

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Related citations in PubMed

KOBAS server: a web based platform for automated annotation ; [Nucleic Acids Res. 2006]

IPAD: the Integrated Pathway Analysis Database for Systematic Enrich [BMC Bioinformatics. 2012]

GDA server: TST based digital gene expression profiling. [Nucleic Acids Res. 2005]

MyBioNet: interactively visualize, edit and merge biological networks on the <div>Bioinformatics. 2014</div>

NCBI-BLAST

<http://blast.ncbi.nlm.nih.gov/Blast.cgi>

 **BLAST®** Basic Local Alignment Search Tool

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BLAST finds regions of similarity between biological sequences. [more...](#)

New DELTA-BLAST, a more sensitive protein-protein search [Go](#)

BLAST Assembled RefSeq Genomes

Choose a species genome to search, or [list all genomic BLAST databases](#).

Human	Oryza sativa	Gallus gallus
Mouse	Bos taurus	Pan troglodytes
Rat	Danio rerio	Microbes
Arabidopsis thaliana	Drosophila melanogaster	Apis mellifera

Basic BLAST

Choose a BLAST program to run.

nucleotide blast	Search a nucleotide database using a nucleotide query <i>Algorithms: blastn, megablast, discontinuous megablast</i>
protein blast	Search protein database using a protein query <i>Algorithms: blastp, psi-blast, phi-blast, delta-blast</i>
blastx	Search protein database using a translated nucleotide query
tblastn	Search translated nucleotide database using a protein query
tblastx	Search translated nucleotide database using a translated nucleotide query

Specialized BLAST

Your Recent Results **New!**

[All Recent results...](#)

News

[Update to organism BLAST databases](#)

The organism BLAST pages are being updated to use top-level (chromosome + unplaced and unlocalized scaffolds) RefSeq genomic records instead of scaffold records.

Thu, 17 Oct 2013 14:00:00 EST

[More BLAST news...](#)

Tip of the Day

[Use Genomic BLAST to see the genomic context](#)

If you are interested in the evolution of a particular gene or gene family it is often interesting to examine the intro-exon structure even across species.

[More tips](#)

NCBI-BLAST

<http://blast.ncbi.nlm.nih.gov/Blast.cgi>

- Online: NCBI-BLAST(blast.ncbi.nlm.nih.gov/Blast.cgi)
- Standalone: BLAST+
- Embedded in webpage: `wwwblast`

EBI (<http://www.ebi.ac.uk/services>)

EMBL-EBI

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XXX DNA & RNA

genes, genomes & variation

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RNA, protein & metabolite expression

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Systems

reactions, interactions & pathways

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chemogenomics & metabolomics

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ArrayExpress

ChEMBL

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Support

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Guide to resources

Service news

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Databases

**Comprehensive
annotation**

Ensembl

Sequence archive

**ENA, EGA, EBI
Metagenomics**

**Large-scale
sequencing project**

1000 Genomes

Tools

Sequence alignment

**BLAST,
Clustal Omega**

Expression

Databases

**Expression profile
archive**

**ArrayExpress,
Expression Atlas**

Proteins

Databases

Protein annotation

**UniProt, InterPro,
Pfam**

**Proteomics data
archive**

PRIDE

Protein structure

**PDBe, PDBsum,
Quaternary structure**

Tools

**Protein function
prediction**

InterProScan

**Protein structure
prediction**

PDBePISA

**Protein function
prediction**

ProFunc

**Protein structure
alignment**

PDBeFold

Reactions, pathways, ontologies

Interactions, pathways & networks

Databases

Molecular interaction

IntAct

Pathway database

Reactome

Network dynamics

BioModels

Small molecules

Databases

**Annotations of
chemicals**

ChEBI, ChEMBL

Metabolomics archive

Metabolights

Enzymes and reactions

Databases

Enzyme annotation

**EBI Enzyme Portal,
Catalytic Site Atlas**

Ontologies

Databases

Ontologies

**Gene Ontology, Systems
Biology Ontology,
Experimental Factor
Ontology**

Tools

**Search engine for
ontologies**

**Ontology Lookup Service,
QuickGO**

Ensembl (http://ensembl.org)

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Search: for
e.g. BRCA2 or rat X:100000..200000 or coronary heart disease

Browse a Genome

The Ensembl project produces genome databases for vertebrate and other eukaryotic species, and makes this information freely available online.

Popular genomes

 **Human**
GRCh37

 **Mouse**
NCBI/mm10

 **Zebrafish**
GRCz10

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All genomes

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Other species are available in [Ensembl Fungi](#) and [Ensembl Genomes](#)

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Variant Effect Predictor



Gene expression in different tissues



Find SNPs and other variants for my gene



Retrieve gene sequence



Compare genes across species



Use my own data in Ensembl



Learn about a disease or phenotype



What's New in Release 73 (September 2013)

- [New search and analysis tools](#)
- [New species: Duck \(Anas platyrhynchos\) and Pycnodon \(Pycnodon olivaceus\)](#)
- [Updated patches for the human assembly \(GRCh37.p12\)](#)

[Full details of this release](#)
[More release news on our blog](#)

Latest blog posts

- 19 Nov 2013: [Ensembl Recombinase](#)
- 01 Nov 2013: [Upgrade of public MySQL servers](#)
- 01 Nov 2013: [Redirection of archive 60](#)

[Go to Ensembl blog](#)

Did you know...?



If you're using the VEP script, please use the newer. Our Variant Effect Predictor tool also has a [web](#) interface.



Ensembl is a joint project between [EMBL - EBI](#) and the [Wellcome Trust Sanger Institute](#) to develop a software system which produces and maintains automatic annotation on selected eukaryotic genomes. Ensembl receives major funding from the Wellcome Trust. Our [acknowledgements page](#) includes a list of additional current and previous funding bodies.



Released version 73 - September 2013 to [WIG](#) / [EBI](#)

[Permanent link](#) - [View in archive site](#)

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Lots of Species in Ensembl

 Aardvark (preview - assembly only) <i>Orycteropus afer afer</i> OryAfe1	 Gibbon <i>Nomascus leucogenys</i> Nleu1.0	 Platypus <i>Ornithorhynchus anatinus</i> OANAS
 Alpaca <i>Vicugna pacos</i> vicPac1	 Gorilla <i>Gorilla gorilla gorilla</i> gorGor3.1	 Prairie vole (preview - assembly only) <i>Microtus ochrogaster</i> MicOch1.0
 Anole lizard <i>Anolis carolinensis</i> AnoCar2.0	 Guinea Pig <i>Cavia porcellus</i> cavPor3	 Rabbit <i>Oryctolagus cuniculus</i> OryCun2.0
 Armadillo (preview new assembly DasNov3.0) <i>Dasypus novemcinctus</i> dasNov2	 Hedgehog <i>Erinaceus europaeus</i> HEDGEHOG	 Rat <i>Rattus norvegicus</i> Rnor_5.0
 Baboon (preview - assembly only) <i>Papio hamadryas</i> Pham	 Horse <i>Equus caballus</i> EquCab2	 Rhinoceros (preview - assembly only) <i>Ceratotherium simum simum</i> CerSimSim1
 Budgerigar (preview - assembly only) <i>Melopsittacus undulatus</i> MelUnd6.3	 Human <i>Homo sapiens</i> GRCh37	 Saccharomyces cerevisiae <i>Saccharomyces cerevisiae</i> EF4
 Bushbaby <i>Otolemur garnettii</i> OtoGar3	 Hyrax <i>Procavia capensis</i> proCap1	 Sheep (preview - assembly only) <i>Ovis aries</i> Oar_v3.1
 Clonintestinalis <i>Clonintestinalis</i> KH	 Kangaroo rat <i>Dipodomys ordii</i> dipOrd1	 Shrew (preview new assembly SorAra2.0) <i>Sorex araneus</i> COMMON_SHREW1
 Clon savignyi <i>Clon savignyi</i> CSAV2.0	 Lamprey <i>Petromyzon marinus</i> Pmarinus_7.0	 Sloth <i>Choloepus hoffmanni</i> choHof1
 Caenorhabditis elegans <i>Caenorhabditis elegans</i> WBcel235	 Lesser hedgehog tenrec <i>Echinops telfairi</i> TENREC	 Spotted Gar (preview - assembly only) <i>Lepisosteus oculatus</i> LepOcu1
 Cat <i>Felis catus</i> Felis_catus_6.2	 Macaque <i>Macaca mulatta</i> MMUL_1	 Squirrel <i>Sciurus harrisi</i> spet12
 Cave fish (preview - assembly only) <i>Astyanax mexicanus</i> AstMex102	 Marmoset <i>Callithrix jacchus</i> C_jacchus3.2.1	 Squirrel monkey (preview - assembly only) <i>Saimiri boliviensis</i> SaiBol1.0

Four Types of Data in Ensembl

- **Curated/Reference:** Data that are manually curated based on experimental results and validations. They serve as the “reference” for future studies.
- **Large-scale projects:** Data that are not curated but are produced by high-throughput experiments. They tend to have a large sample size or data volume.
- **Other studies:** Low-throughput experimental data that are not curated, possibly with some computational postprocessing.
- **Only computationally analyzed:** Secondary data that are generated by running automated algorithms over curated/reference data and large-scale high-throughput data (e.g. the genome sequence).

UniProtKB (<http://www.uniprot.org/>)

UniProtKB - UNIPROT

Search | Basic | Align | Retrieve | ID Mapping

Search in:

1 - 25 of 573 results for cxcl in UniProtKB sorted by score descending

Download | Browse by taxonomy, keyword, gene ontology, enzyme class or pathway | Reduce sequence redundancy to 100%, 90% or 50%

Page 1 of 23 | Next

Results [Customize](#)

Show only reviewed (153) (UniProtKB/Swiss-Prot) or unreviewed (158) (UniProtKB/TrEMBL) entries

Restrict term 'cxcl' to gene name (158), protein name (170), strain (11), taxonomy (11), web resource (14)

Entry	Entry name	Status	Protein names	Gene names	Organism	Length
Q1RA3U8	CXL18_RAT	★	C-X-C motif chemokine 18	Cxcl18	Rattus norvegicus (Rat)	247
Q1R8U2	CXL18_MOUSE	★	C-X-C motif chemokine 18	Cxcl18 Sypha	Mus musculus (Mouse)	246
Q22RT9	CXL18_BOVIN	★	C-X-C motif chemokine 18	CXCL18	Bos taurus (Bovine)	251
Q1D3P2	CXL18_PIG	★	C-X-C motif chemokine 18	CXCL18	Sus scrofa (Pig)	250
P07778	CXL18_HUMAN	★	C-X-C motif chemokine 18	CXCL18 INP10 XLYK10	Homo sapiens (Human)	88
Q9H2A7	CXL18_HUMAN	★	C-X-C motif chemokine 18	CXCL18 SCY815 SRP50X UNQ2759/PRO5714	Homo sapiens (Human)	251
P10730	PR4V_HUMAN	★	Platelet factor 4 variant	PR4V1 CXCL18V XLYK1V	Homo sapiens (Human)	104
Q5U2W9	Q5U2W9_MOUSE	★	Cxcl1 protein	Cxcl1	Mus musculus (Mouse)	107
Q8R392	Q8R392_MOUSE	★	Cxcl11 protein	Cxcl11	Mus musculus (Mouse)	100
Q1HACD	Q1HACD_MOUSE	★	Cxcl14 protein	Cxcl14	Mus musculus (Mouse)	81
Q0P519	Q0P519_BOVIN	★	CXCL12 protein	CXCL12	Bos taurus (Bovine)	88
Q0K453	Q0K453_RAT	★	BRAX	Cxcl14	Rattus norvegicus (Rat)	90
K09Y17	K09Y17_RABIT	★	CXCL13	CXCL13	Oryctolagus cuniculus (Rabbit)	88
Q6PUJ1	Q6PUJ1_PIG	★	CXCL2	CXCL2	Sus scrofa (Pig)	107
Q5ID16	Q5ID16_RAT	★	Chemokine (C-X-C motif) ligand 13	Cxcl13 LOC420335	Rattus norvegicus (Rat)	105
A1ZPH8	A1ZPH8_DANRE	★	Chemokine CXCL-C5c	cxcl-c5c	Danio rerio (Zebrafish) (Brachydanio rerio)	147
Q1K481	Q1K481_RAT	★	Chemokine (C-X-C motif) ligand 8	Cxcl8 Mig CG-6848b	Rattus norvegicus (Rat)	126
Q02YV0	Q02YV0_RAT	★	Protein Cxcl12	Cxcl12	Rattus norvegicus (Rat)	112
Q1RAV10	Q1RAV10_DANRE	★	Chemokine ligand 12	cxcl12a	Danio rerio (Zebrafish) (Brachydanio rerio)	93
P10228	CXCL8_MOUSE	★	C-X-C motif chemokine 8	Cxcl8 Sypha	Mus musculus (Mouse)	142
Q55218	Q55218_TUPAE	★	CXCL9	CXCL9	Tupaia belangeri (Common tree shrew) (Tupaia glis)	92

IntAct (<http://www.ebi.ac.uk/intact/>)

The screenshot displays the IntAct website interface. At the top, there is a navigation bar with links for Services, Research, Training, and About us. The IntAct logo is prominently displayed. On the left side, a sidebar contains links to Home, Advanced Search, Tools, Data Submission, Downloads, Documentation, Acknowledgements, and Contact Us. Below these links, there is a Twitter feed and a link to the IMEx homepage. The main content area features a search bar with a 'Search' button and a 'Clear' button. Below the search bar, there are instructions on how to use the search function and a list of examples. A navigation bar below the search area includes links for Home, Search, Interactions (428711), Browse, Lists, Interaction Details, Molecular View, and Graph. The main content area also includes a section for 'Citing IntAct' with a list of references and a 'Dataset of the month: November' section. On the right side, there is a section for 'Manually curated content is added to IntAct by curators at the EMBL-EBI and the following organisations:' with logos for MINT, UniProt, SLB, I2D, InnateDB, MINTACT, MatrixDB, and MB:Info.

EMBL-EBI

Services Research Training About us

IntAct

Home Advanced Search Tools Data Submission Downloads Documentation Acknowledgements Contact Us

IntAct v4.1.1

Twitter

Twitter by @intact_project

Back to IntAct homepage

IMEx member

IntAct v4.1.1

Search: Search Clear Show Advanced Fields MINT syntax reference

Examples

- Gene name: e.g. RPL62
- UniProt ID: e.g. Q05522
- UniProt ID: e.g. Q05522
- UniProt ID: e.g. Q05522

Interactions (428711)

Publications Experiments Interactions Interactors

12332 21582 138711 75810

Citing IntAct

- The MINT project—IntAct as a common curation platform for 11 molecular interaction databases. Orchard S et al. (2013) 243544511. Nucleic Acids Res. (2013) doi:10.1093/nar/gkt116 [Abstract] [Full Text]

Dataset of the month: November

- Identification and Comparative Analysis of Hepatitis C Virus-Host Cell Protein Interactions.

Manually curated content is added to IntAct by curators at the EMBL-EBI and the following organisations:

MINT MINT

UniProt UniProt

SLB SLB

I2D I2D

InnateDB InnateDB

MINTACT MINTACT

MatrixDB MatrixDB

MB:Info MB:Info

Clustal Omega (<http://www.clustal.org/omega/>)

EMBL-EBI  Services Research Training About us

Clustal Omega

Input form Web services Help & Documentation [Share](#) [Feedback](#)

[Tools](#) > [Multiple Sequence Alignment](#) > Clustal Omega

Multiple Sequence Alignment

Clustal Omega is a new multiple sequence alignment program that uses seeded guide trees and HMM profile-profile techniques to generate alignments.

STEP 1 - Enter your input sequences

Enter or paste a set of sequences in any supported format:

Or, upload a file: 選択されていません

STEP 2 - Set your parameters

OUTPUT FORMAT

The default settings will fulfil the needs of most users and, for that reason, are not visible.
 (Click here, if you want to view or change the default settings.)

STEP 3 - Submit your job

☐ Be notified by email (Tick this box if you want to be notified by email when the results are available)

If you plan to use these services during a course please [contact us](#).

Please read the [FAQ](#) before seeking help from our support staff.

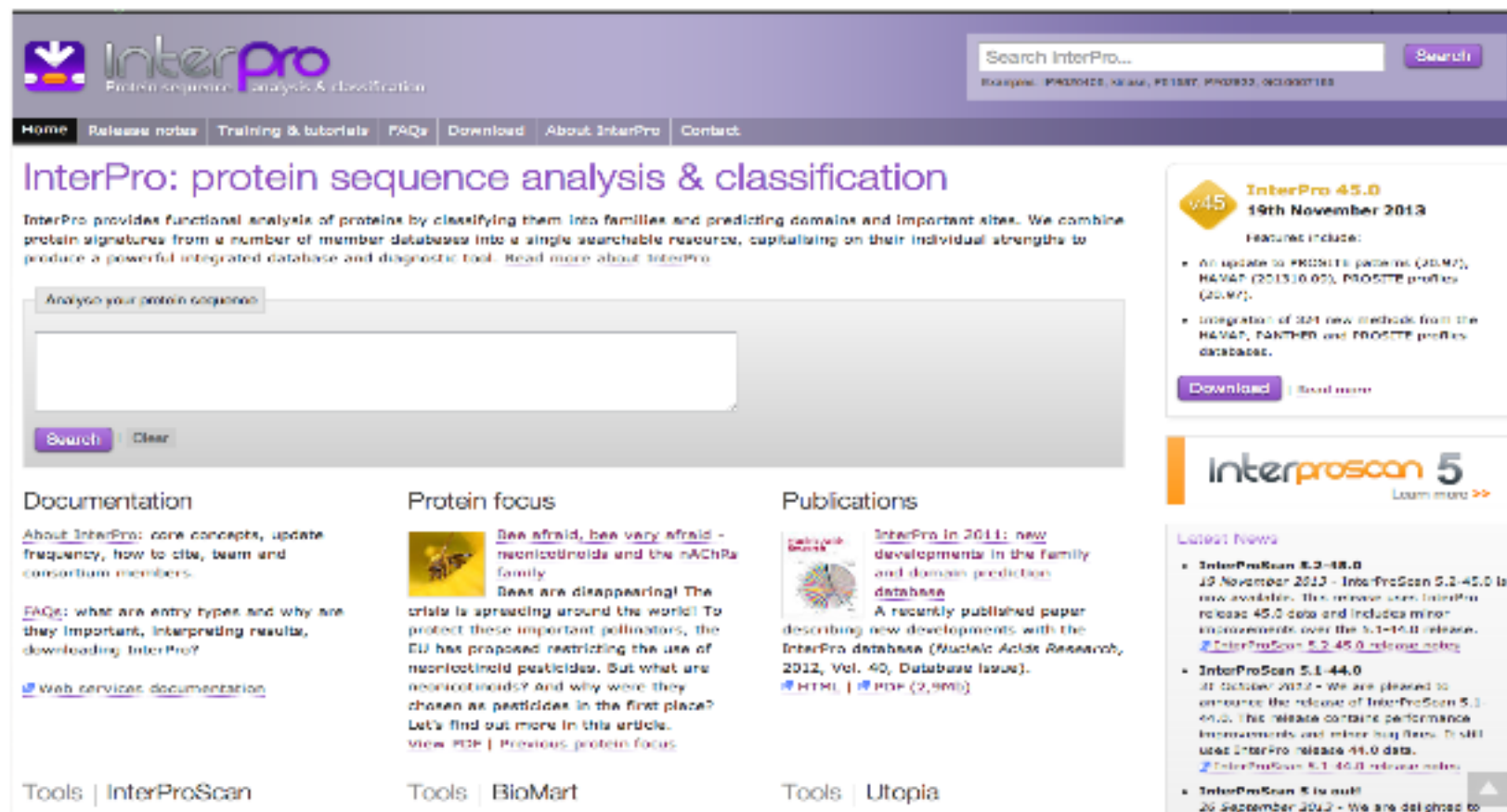
ClustalW2 – Phylogeny

(http://www.ebi.ac.uk/Tools/phylogeny/clustalw2_phylogeny/)

The screenshot shows the ClustalW2 - Phylogeny web interface. At the top is the EMBL-EBI logo and navigation links: Services, Research, Training, and About us. Below this is a search bar. The main header is "ClustalW2 - Phylogeny". Underneath the header are links for "Input form", "Web services", and "Help & Documentation", along with "Share" and "Feedback" buttons. The breadcrumb trail reads "Tools > Phylogeny > ClustalW2". The section title "Phylogeny" is followed by the text "Commonly used phylogenetic tree generation methods provided by the ClustalW2 program." The interface is divided into two steps. "STEP 1 - Enter your multiple sequence alignment" contains a text area for pasting a multiple sequence alignment in any supported format, and a file upload option with a "Choose..." button. "STEP 2 - Set your Phylogeny options" includes a table of settings:

TREE FORMAT	DISTANCE CORRECTION	EXCLUDE GAPS	CLUSTERING METHOD	P.T.M.

InterPro & InterProScan (http://www.ebi.ac.uk/interpro/)



The screenshot displays the InterPro website interface. At the top, there is a search bar with the text "Search InterPro..." and a "Search" button. Below the search bar is a navigation menu with links: Home, Release notes, Training & tutorials, FAQs, Download, About InterPro, and Contact. The main heading is "InterPro: protein sequence analysis & classification". Below this, a brief description states: "InterPro provides functional analysis of proteins by classifying them into families and predicting domains and important sites. We combine protein signatures from a number of member databases into a single searchable resource, capitalising on their individual strengths to produce a powerful integrated database and diagnostic tool. [Read more about InterPro](#)".

Below the description is a section for "Analyse your protein sequence" with a large text input field and "Search" and "Clear" buttons. To the right of this section is a box for "InterPro 45.0" dated "19th November 2013". It lists features: "An update to PROSITE patterns (20.M7), HAVAT (2013.0.05), PROSITE profiles (20.M7)" and "Integration of 321 new methods from the HAVAT, PANTHER and PROSITE profiles databases." Below this are "Download" and "Read more" links.

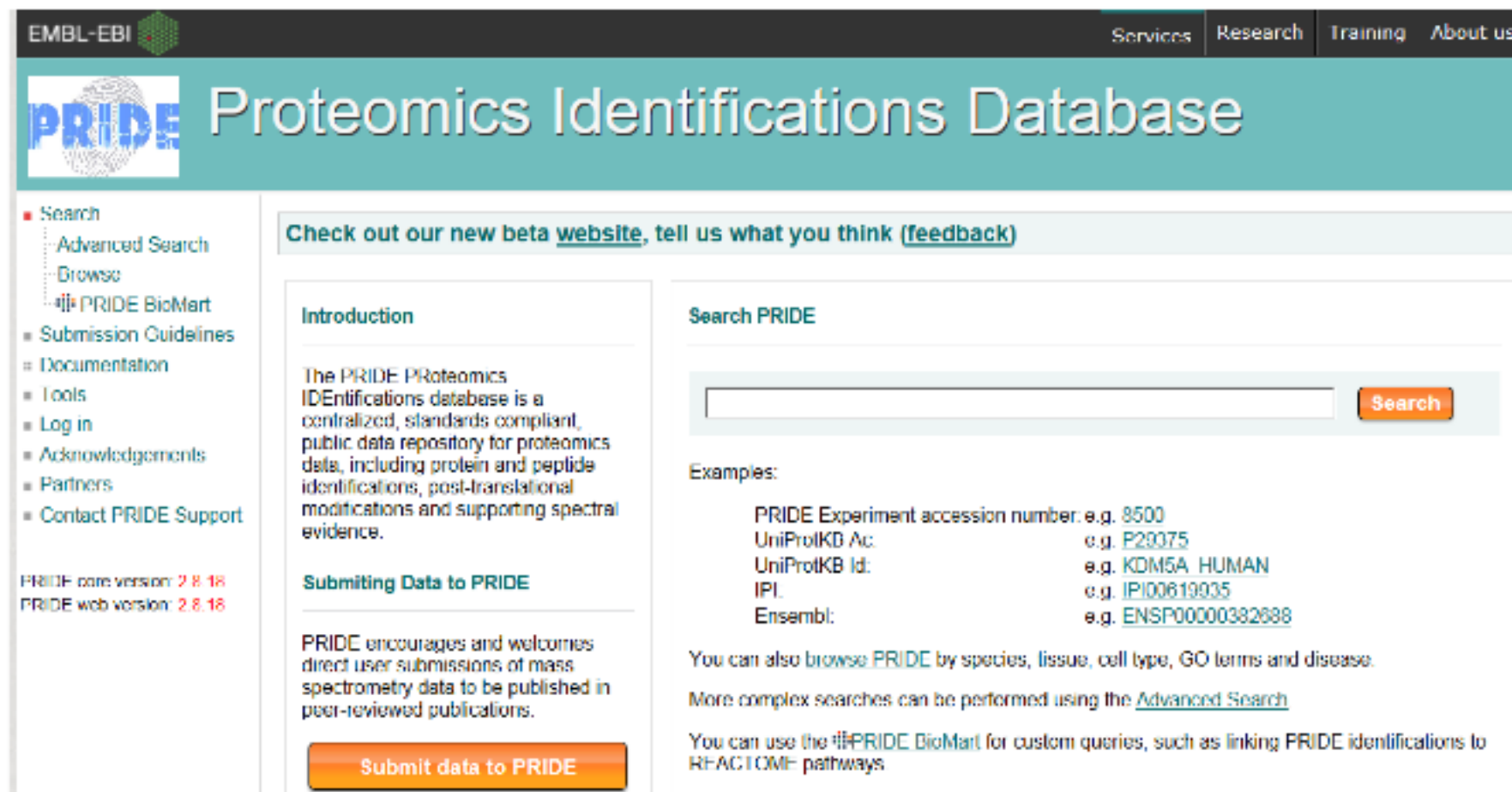
Below the main heading are three columns of content:

- Documentation**: "About InterPro: core concepts, update frequency, how to cite, team and consortium members." and "FAQs: what are entry types and why are they important, interpreting results, downloading InterPro?". A link for "Web services documentation" is also present.
- Protein focus**: A section titled "Bees afraid, bees very afraid - neonicotinoids and the nAChRα family". It discusses the decline of bees and the use of neonicotinoid pesticides. It includes a "View PDF" link and a "Previous protein focus" link.
- Publications**: A section titled "InterPro in 2011: new developments in the family and domain prediction database". It mentions a recently published paper describing new developments with the InterPro database (Nucleic Acids Research, 2012, Vol. 40, Database Issue). It includes "HTML" and "PDF (2.9Mb)" links.

At the bottom, there are three tool links: "Tools | InterProScan", "Tools | BioMart", and "Tools | Utopia". On the right side, there is a section for "Interproscan 5" with a "Learn more" link. Below that is a "Latest News" section with three items:

- InterProScan 5.2-45.0**: "19 November 2013 - InterProScan 5.2-45.0 is now available. This release uses InterPro release 45.0 data and includes minor improvements over the 5.1-45.0 release. [InterProScan 5.2-45.0 release notes](#)"
- InterProScan 5.1-44.0**: "31 October 2012 - We are pleased to announce the release of InterProScan 5.1-44.0. This release contains performance improvements and minor bug fixes. It still uses InterPro release 44.0 data. [InterProScan 5.1-44.0 release notes](#)"
- InterProScan 5 is out!**: "26 September 2012 - We are delighted to"

PRIDE (<http://www.ebi.ac.uk/pride/>)



The screenshot shows the PRIDE Proteomics Identifications Database homepage. At the top, there is a navigation bar with links for Services, Research, Training, and About us. Below this, the PRIDE logo is displayed next to the title 'Proteomics Identifications Database'. A sidebar on the left contains a search menu with options like Advanced Search, Browse, PRIDE BioMart, Submission Guidelines, Documentation, Tools, Log in, Acknowledgements, Partners, and Contact PRIDE Support. The main content area features a banner for a new beta website, an introduction to the database, a search bar with a 'Search' button, and a section for submitting data to PRIDE. The introduction states that the PRIDE Proteomics IDentifications database is a centralized, standards compliant, public data repository for proteomics data, including protein and peptide identifications, post-translational modifications, and supporting spectral evidence. The search section provides examples of search terms: PRIDE Experiment accession number (e.g. 8500), UniProtKB Ac. (e.g. P29375), UniProtKB Id. (e.g. KDM5A_HUMAN), IPI. (e.g. IPI00619935), and Ensembl (e.g. ENSP00000382888). It also mentions that users can browse PRIDE by species, tissue, cell type, GO terms, and disease, and that more complex searches can be performed using the Advanced Search. A final note mentions the PRIDE BioMart for custom queries, such as linking PRIDE identifications to REACTOME pathways.

EMBL-EBI

Services Research Training About us

PRIDE Proteomics Identifications Database

■ Search

- Advanced Search
- Browse
- PRIDE BioMart
- Submission Guidelines
- Documentation
- Tools
- Log in
- Acknowledgements
- Partners
- Contact PRIDE Support

PRIDE core version: 2.8.18
PRIDE web version: 2.8.18

Check out our new beta [website](#), tell us what you think ([feedback](#))

Introduction

The PRIDE Proteomics IDentifications database is a centralized, standards compliant, public data repository for proteomics data, including protein and peptide identifications, post-translational modifications and supporting spectral evidence.

Submitting Data to PRIDE

PRIDE encourages and welcomes direct user submissions of mass spectrometry data to be published in peer-reviewed publications.

[Submit data to PRIDE](#)

Search PRIDE

[Search](#)

Examples:

PRIDE Experiment accession number: e.g. 8500
UniProtKB Ac. e.g. P29375
UniProtKB Id. e.g. KDM5A_HUMAN
IPI. e.g. IPI00619935
Ensembl: e.g. ENSP00000382888

You can also browse PRIDE by species, tissue, cell type, GO terms and disease.

More complex searches can be performed using the [Advanced Search](#)

You can use the PRIDE BioMart for custom queries, such as linking PRIDE identifications to REACTOME pathways

UCSC Genome Bioinformatics (<http://genome.ucsc.edu/>)

UNIVERSITY OF CALIFORNIA
SANTA CRUZ

Genomics
Institute



Genome Browser

GenomesGenome BrowserToolsMirrorsDownloadsMy DataProjectsHelpAbout Us



Search genes, data, help docs and more...Search

Tools



hg38



hg19



mm39

- **Genome Browser** - Interactively visualize genomic data
- **BLAT** - Rapidly align sequences to the genome
- **In-Silico PCR** - Rapidly align PCR primer pairs to the genome
- **Table Browser** - Download and filter data from the Genome Browser
- **LiftOver** - Convert genome coordinates between assemblies
- **REST API** - Returns data requested in JSON format
- **Variant Annotation Integrator** - Annotate genomic variants
- **More tools...**

News

- Mar. 07, 2024 - **New Prediction Scores super track and BayesDel track for hg19**
- Mar. 05, 2024 - **New JASPAR tracks: Human (hg19/hg38) - Mouse (mm10/mm39)**
- Mar. 01, 2024 - **AbSplice Prediction Scores for hg38**
- Feb. 21, 2024 - **New DECIPHER Dosage Sensitivity tracks for Human (hg19/hg38)**
- Feb. 14 2024 - **New GENCODE gene tracks: V45 (hg19/hg38) - VM34 (mm39)**
- Feb. 12, 2024 - **Variants of Concern SARS-CoV-2 track updated with Omicron vari**

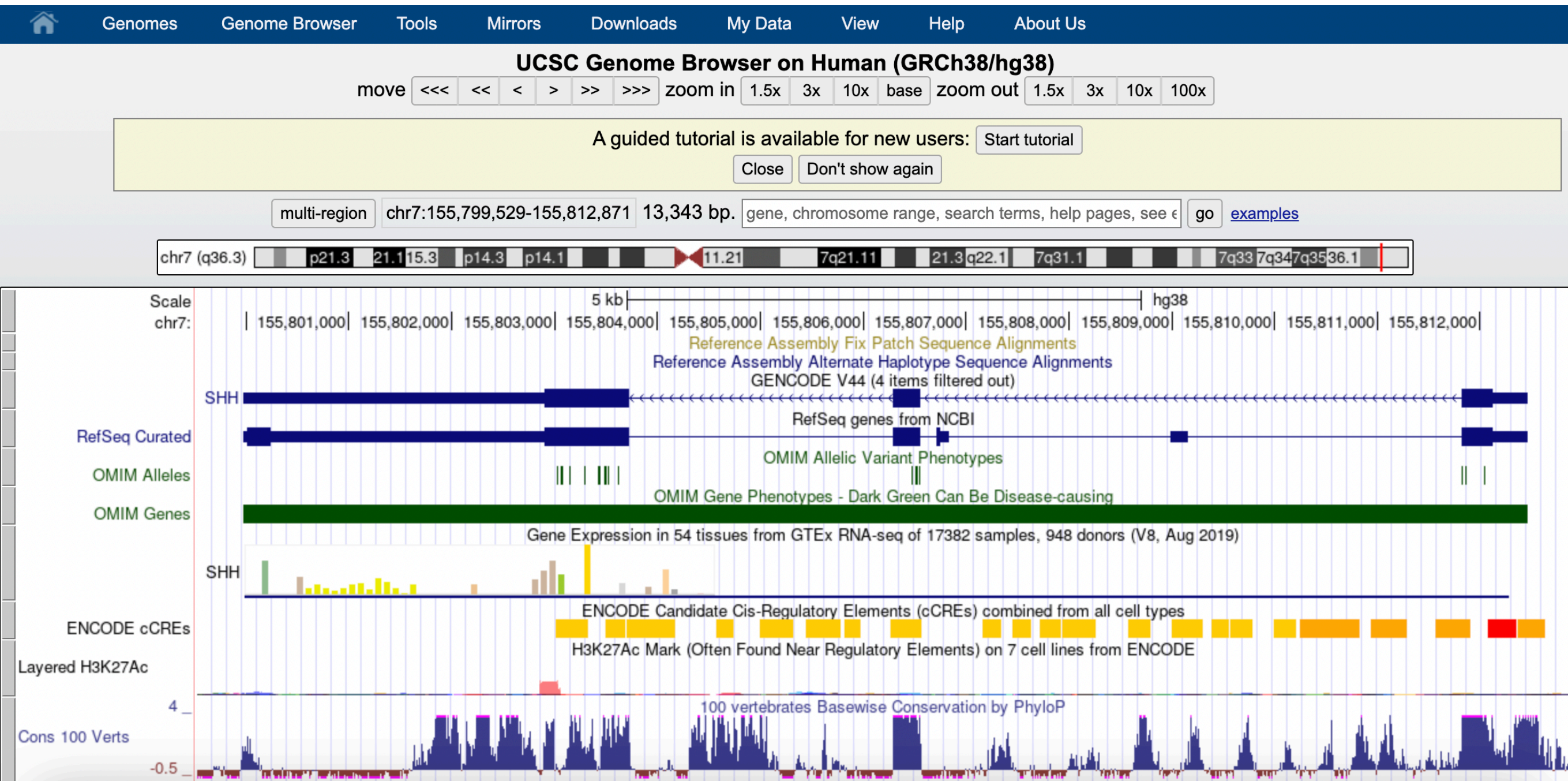
More news...

Subscribe



UCSC Genome Browser

Browser (<http://genome.ucsc.edu/>)





genome.gov

National Human Genome Research Institute

National Institutes of Health

Google™ Search

SEARCH

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Research at NHGRI

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Issues in Genetics

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Careers & Training

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Home > Research Funding > Research Funding Divisions > Division of Genome Sciences > ENCODE Project

Division of Genome Sciences

Centers of Excellence in Genomic Science ▶

Division Staff ▶

ENCODE Project ▶

Functional Analysis Program

Genetic Variation Program ▶

Genome Informatics and Computational Biology Program ▶

Genome Technology Program ▶

NHGRI Genome Sequencing Program (GSP) ▶

The ENCODE Project: ENCyclopedia Of DNA Elements

Share Print



- ① [Overview](#)
- ① [Publications, Features and Press Releases](#)
- ① [Consortium Membership](#)
- ① [Data Release Policy](#)
- ① [Accessing ENCODE Data](#)
- ① [ENCODE Tutorials](#)
- ① [Common Cell Types](#)
- ① [Requests for Application \(RFAs\)](#)
- ① [Program Staff](#)

Follow the ENCODE Project on:

[Facebook](#)

[Twitter](#)

See Also:

[Identification and analysis of functional elements in 1% of the human genome by the ENCODE pilot project](#)
Nature, June 13, 2007

[Major Findings from The ENCODE Pilot Project](#)
June 2007

[The modENCODE Project](#)

[Grants Home](#)

On Other Sites:

[The ENCODE \(ENCyclopedia Of DNA Elements\) Project](#)

ENCODE data portal at UCSC

(<http://genome.ucsc.edu/encode/>)



Encyclopedia of DNA Elements

General

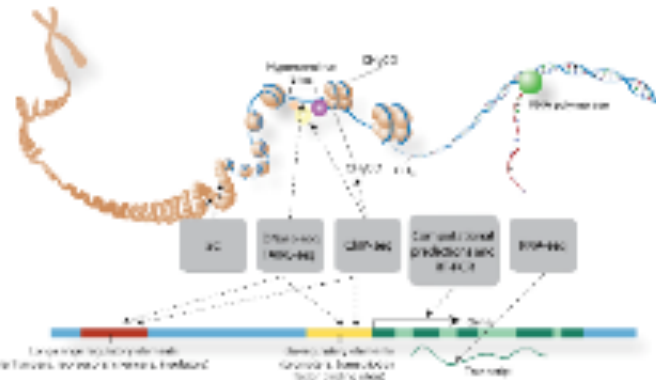
- Resources & FAQ
- Publications
- Software Tools
- Data Standards

Human

- Downloads
- Experiment Matrix
- Search
- Genome Browser (hg19)
- Integrative Analysis

About ENCODE Data

The [Encyclopedia of DNA Elements](#) (ENCODE) Consortium is an international collaboration of research groups funded by the National Human Genome Research Institute (NHGRI). The goal of ENCODE is to build a comprehensive parts list of functional elements in the human genome, including elements that act at the protein and RNA levels, and regulatory elements that control cells and circumstances in which a gene is active.



ENCODE data are now available for the entire human genome. **All ENCODE data are free and available for immediate use via :**

- [Search](#) for displayable tracks and downloadable files
- [Download](#) of data files
- [Visualization](#) in the UCSC Genome Browser (ENCODE data marked with the NHGRI logo)
- [Data mining](#) with the UCSC Table Browser and other [UCSC Genome Bioinformatics tools](#)

To search for ENCODE data related to your area of interest and set up a browser view, use the UCSC [Experiment Matrix](#) or [Track Search tool](#) (Advanced features). The [Experiment List \(Human\)](#) and [Experiment List \(Mouse\)](#) links provide comprehensive listings of ENCODE data that is released or in preparation.

UCSC *In-Silico* PCR (<http://genome.ucsc.edu/cgi-bin/hgPcr>)

 Genomes Genome Browser Tools Mirrors Downloads My Data About Us Help

UCSC In-Silico PCR

Genome:

Human

Assembly:

Feb. 2009 (GRCh37/hg19)

Target:

genome assembly

Forward Primer:

Reverse Primer:

submit

Max Product Size:

4000

Min Perfect Match:

15

Min Good Match:

15

Flip Reverse Primer: ☐

About In-Silico PCR

In-Silico PCR searches a sequence database with a pair of PCR primers, using an indexing strategy for fast performance.

Configuration Options

Genome and Assembly - The sequence database to search.

Target - If available, choose to query transcribed sequences.

Forward Primer - Must be at least 15 bases in length.

Reverse Primer - On the opposite strand from the forward primer. Minimum length of 15 bases.

Max Product Size - Maximum size of amplified region.

Min Perfect Match - Number of bases that match exactly on 3' end of primers. Minimum match size is 15.

Min Good Match - Number of bases on 3' end of primers where at least 2 out of 3 bases match.

Flip Reverse Primer - Invert the sequence order of the reverse primer and complement it.

Output

When successful, the search returns a sequence output file in fasta format containing all sequence in the database that lie between and include the primer pair. The fasta header describes the region in the database and the primers. The fasta body is capitalized in areas where the primer sequence matches the database sequence and in lower-case elsewhere. Here is an example from human:

```
>chr22:31000551+31001000 TAACAGATTGATGATGCATGAAATGGG CCCATGAGTGGCTCCTAAAGCAGCTGC
TtACAGATTGATGATGCATGAAATGGGgggtggcaggggtgggggtga
```


PDB (<http://www.rcsb.org/>)

The screenshot shows the Protein Data Bank (PDB) website homepage. At the top, there is a navigation bar with the PDB logo and links to "PDB Statistics" and "PDB Data Bank". Below this is a search bar with a dropdown menu showing "Everything", "Author", "Macromolecule", "Sequence", and "Ligand". The search bar contains the text "PDB e.g., PDB ID, molecule name, author". To the left of the search bar, there are links for "Search", "Advanced", and "Browse".

The main content area is titled "Biological Macromolecular Resource" and features a "Full Description" section. This section includes a "Learn: Featured Molecules" area with a "Structural View of Biology" tab. Below this, there is a "Molecule of the Month" section for "SNARE Proteins" and a "Protein Structure Initiative Featured System" section for "Methylation of Arginine".

On the left side, there is a "Customize This Page" section with links to "Available on the App Store", "PDB-101", "HyPDB", "Home", and "Deposition". The "PDB-101" link is highlighted, and it includes a "Full Description" link. The "HyPDB" link is also highlighted, and it includes a "Full Description" link. The "Home" link is highlighted, and it includes a "Full Description" link. The "Deposition" link is highlighted, and it includes a "Full Description" link.

On the right side, there is a "New Features" section with a "Latest release: September 2013" announcement. Below this, there is a "Browse Membrane Transport Proteins" section with a grid of protein structures.

PDB

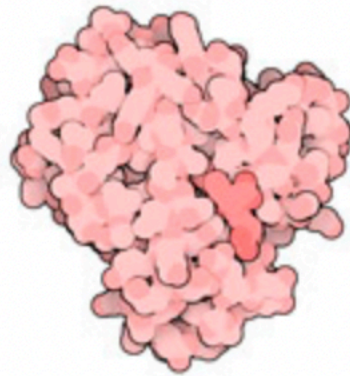
Protein Data Bank (PDB) archive of 3D structure data for biological molecules (proteins, DNA, RNA).

Currently includes > 1TB of structure data, archived world-wide.



The Protein Data Bank, or PDB, is a vital repository for 3D structure data of biological molecules. We'll delve into the significance of PDB, its role in advancing structural biology, and the substantial volume of data it currently archives.

Protein Data Bank in 1973



myoglobin



lamprey
hemoglobin



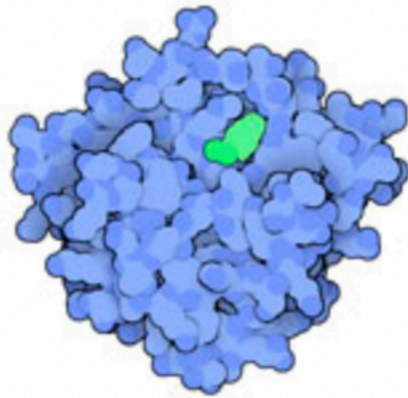
rubredoxin



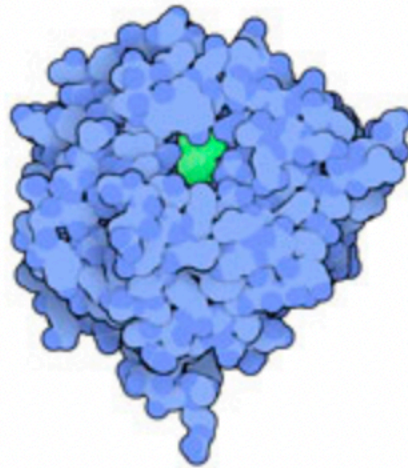
cytochrome b5



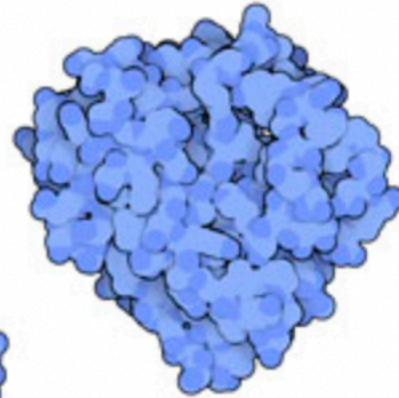
trypsin inhibitor



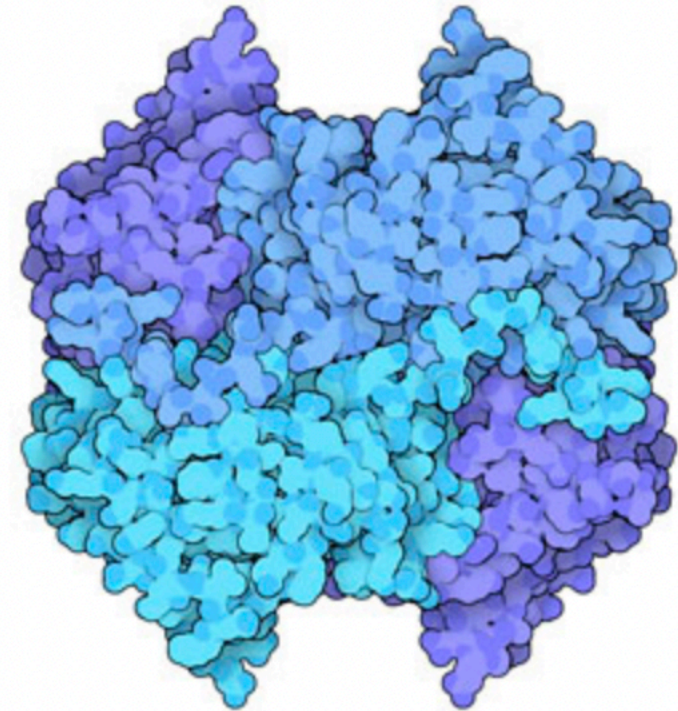
chymotrypsin



carboxypeptidase



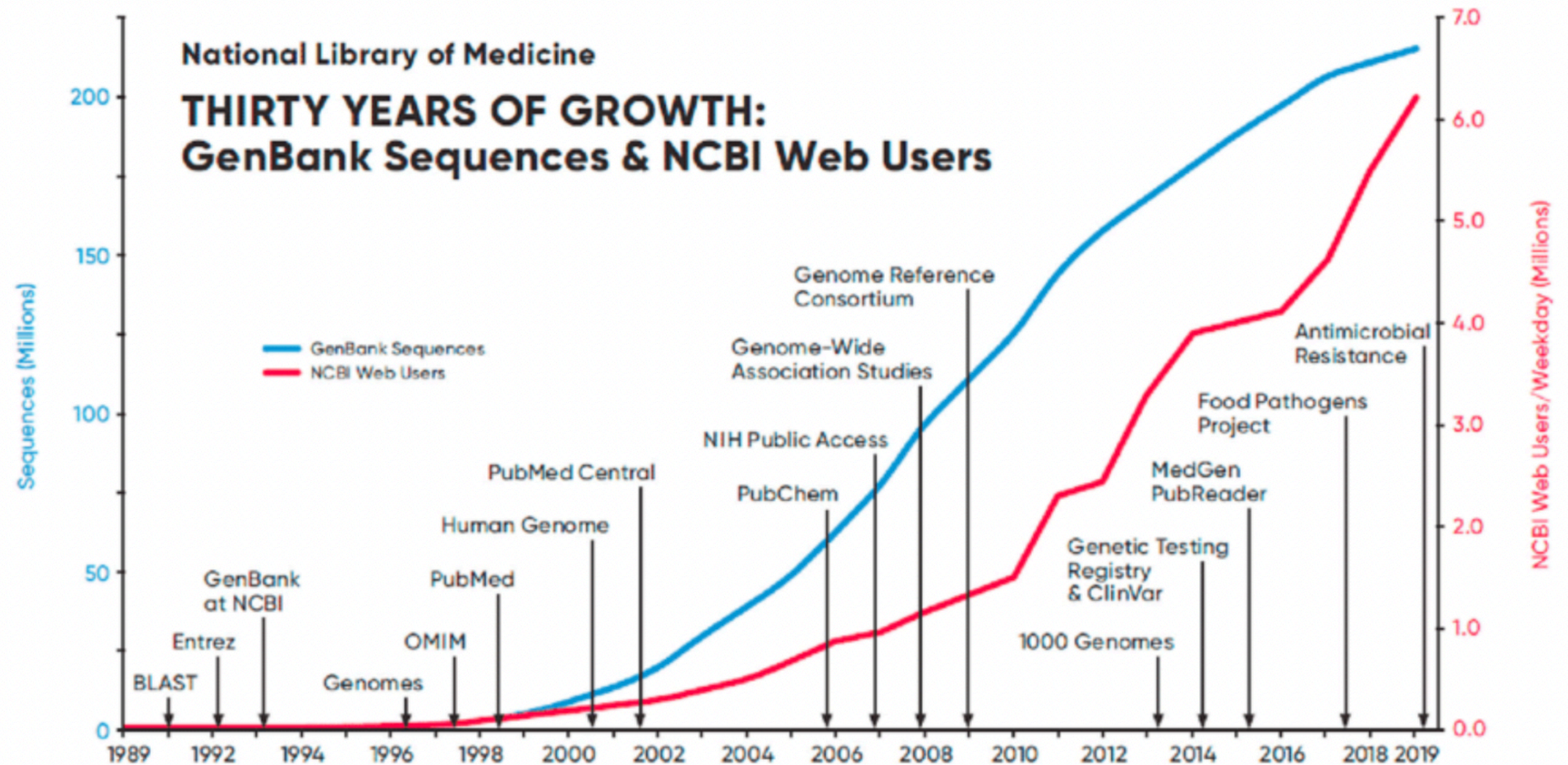
subtilisin



lactate dehydrogenase

GenBank

An annotated collection of all publicly available DNA sequences, which comprises the DNA DataBank of Japan (DDBJ), the European Nucleotide Archive (ENA), and GenBank at NCBI



GenBank stands as a critical resource for DNA sequences. It collaborates with other databases, such as DDBJ and ENA, to provide a comprehensive collection of publicly available DNA sequences.

https://swissmodel.expasy.org/interactive

Start a New Modelling Project

Target
Sequence(s):
*(Format must be
FASTA, Clustal,
plain string, or a valid
UniProtKB AC)*

Paste your target sequence(s) or UniProtKB AC here



 Upload Target Sequence File...

 Validate

Project Title:

Untitled Project

Email:

Optional

Search For Templates

Build Model

Supported Inputs

Sequence(s) ▼

Target-Template Alignment ▼

User Template ▼

DeepView Project ▼

By using the SWISS-MODEL server, you agree to comply with the following [terms of use](#) and to cite the corresponding [articles](#).

You are currently not logged in - to take advantage of the workspace, please [log in](#) or [create an account](#).

(There is no requirement to create an account to use any part of SWISS-MODEL, however you will gain the benefit of seeing a list of your previous modelling projects here.)

SWISS-MODEL Repository

(<http://swissmodel.expasy.org/repository/?pid=smr01&zid=async>)

The SWISS-MODEL Repository is a database of annotated 3D protein structure models generated by the SWISS-MODEL homology-modelling pipeline.

Bienert S, Waterhouse A, de Beer TA, Tauriello G, Studer G, Bordoli L, Schwede T (2017). The SWISS-MODEL Repository - new features and functionality *Nucleic Acids Res.* 45(D1):D313-D319.  [doi>](https://doi.org/10.1093/nar/dkx105)

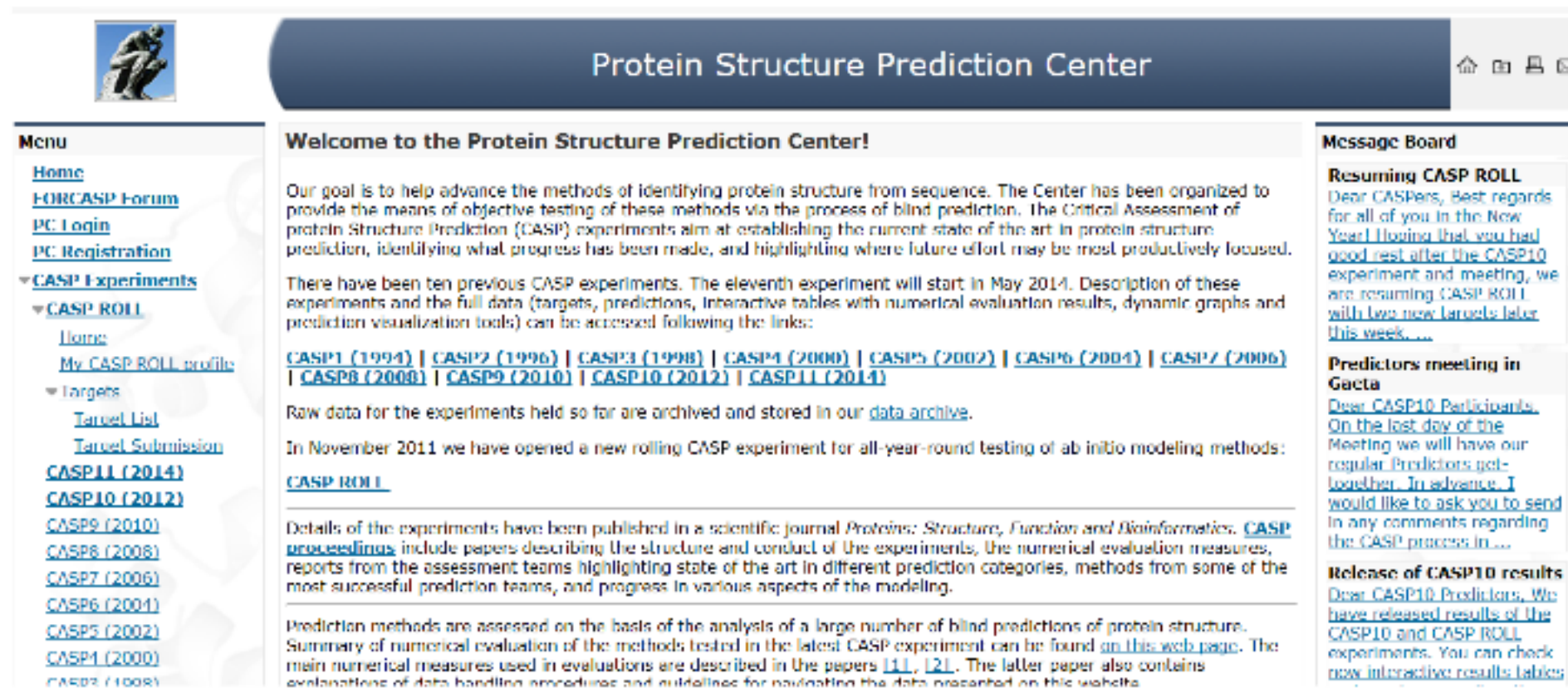
The aim of the SWISS-MODEL Repository is to provide access to an up-to-date collection of annotated 3D protein models generated by automated homology modelling for relevant model organisms and experimental structure information for all sequences in UniProtKB. Regular updates ensure that target coverage is complete, that models are built using the most recent sequence and template structure databases, and that improvements in the underlying modelling pipeline are fully utilised. It also allows users to assess the quality of the models using the latest QMEAN results. If a sequence has not been modelled, the user can build models interactively via the SWISS-MODEL workspace.

Currently the repository contains 2,543,873 models from SWISS-MODEL for UniProtKB targets as well as 203,493 structures from PDB with mapping to UniProtKB.

We currently provide models for the **reference proteomes** of the following model organisms, based on UniProtKB release 2024_01. If you want to download a large number of models, please contact us.

	Proteome Size	Sequences Modelled	Models	Seq Coverage	Metadata (Homology models <i>and</i> experimental structures)	Coordinates (Homology models only)
<i>Homo sapiens</i>	20,597	17,280	42,868		↓ 14.1 MB	↓ 5.5 GB
<i>Mus musculus</i>	21,701	18,785	43,406		↓ 7.6 MB	↓ 3.9 GB
<i>Caenorhabditis elegans</i>	19,827	12,744	22,978		↓ 3.5 MB	↓ 1.6 GB
<i>Escherichia coli</i>	4,403	3,675	6,169		↓ 2.0 MB	↓ 482.5 MB
<i>Arabidopsis thaliana</i>	27,448	20,179	37,235		↓ 5.4 MB	↓ 2.5 GB
<i>Drosophila melanogaster</i>	13,824	10,000	19,691		↓ 3.0 MB	↓ 1.7 GB
<i>Saccharomyces cerevisiae</i>	6,060	4,631	8,516		↓ 2.2 MB	↓ 598.6 MB

CASP (<http://www.predictioncenter.org/>)



The screenshot shows the homepage of the Protein Structure Prediction Center (CASP). The header features a logo of a person climbing a rock and the title "Protein Structure Prediction Center" in a dark blue bar. The main content area is divided into three columns. The left column contains a "Menu" with links to Home, FORCASP Forum, PC Login, PC Registration, and a dropdown for CASP Experiments. The middle column has a "Welcome to the Protein Structure Prediction Center!" section, followed by a paragraph about the center's goal, a list of previous CASP experiments from 1994 to 2014, a link to the data archive, and a notice about a new rolling CASP experiment. The right column contains a "Message Board" with three entries: "Resuming CASP ROLL", "Predictors meeting in Gsta", and "Release of CASP10 results".

Protein Structure Prediction Center

Menu

- [Home](#)
- [FORCASP Forum](#)
- [PC Login](#)
- [PC Registration](#)
- ▼ **CASP Experiments**
 - ▼ **CASP ROLL**
 - [Home](#)
 - [My CASP ROLL profile](#)
 - ▼ **Targets**
 - [Target List](#)
 - [Target Submission](#)
- [CASP11 \(2014\)](#)
- [CASP10 \(2012\)](#)
- [CASP9 \(2010\)](#)
- [CASP8 \(2008\)](#)
- [CASP7 \(2006\)](#)
- [CASP6 \(2004\)](#)
- [CASP5 \(2002\)](#)
- [CASP4 \(2000\)](#)
- [CASP3 \(1998\)](#)

Welcome to the Protein Structure Prediction Center!

Our goal is to help advance the methods of identifying protein structure from sequence. The Center has been organized to provide the means of objective testing of these methods via the process of blind prediction. The Critical Assessment of protein Structure Prediction (CASP) experiments aim at establishing the current state of the art in protein structure prediction, identifying what progress has been made, and highlighting where future effort may be most productively focused.

There have been ten previous CASP experiments. The eleventh experiment will start in May 2014. Description of these experiments and the full data (targets, predictions, interactive tables with numerical evaluation results, dynamic graphs and prediction visualization tools) can be accessed following the links:

[CASP1 \(1994\)](#) | [CASP2 \(1996\)](#) | [CASP3 \(1998\)](#) | [CASP4 \(2000\)](#) | [CASP5 \(2002\)](#) | [CASP6 \(2004\)](#) | [CASP7 \(2006\)](#) | [CASP8 \(2008\)](#) | [CASP9 \(2010\)](#) | [CASP10 \(2012\)](#) | [CASP11 \(2014\)](#)

Raw data for the experiments held so far are archived and stored in our [data archive](#).

In November 2011 we have opened a new rolling CASP experiment for all-year-round testing of ab initio modeling methods: [CASP ROLL](#).

Details of the experiments have been published in a scientific journal *Proteins: Structure, Function and Bioinformatics*. [CASP proceedings](#) include papers describing the structure and conduct of the experiments, the numerical evaluation measures, reports from the assessment teams highlighting state of the art in different prediction categories, methods from some of the most successful prediction teams, and progress in various aspects of the modeling.

Prediction methods are assessed on the basis of the analysis of a large number of blind predictions of protein structure. Summary of numerical evaluation of the methods tested in the latest CASP experiment can be found [on this web page](#). The main numerical measures used in evaluations are described in the papers [\[1\]](#), [\[2\]](#). The latter paper also contains explanations of data handling procedures and guidelines for navigating the data presented on this website.

Message Board

Resuming CASP ROLL
Dear CASPERS, Best regards for all of you in the New Year! Hoping that you had good rest after the CASP10 experiment and meeting, we are resuming CASP ROLL with two new targets later this week....

Predictors meeting in Gsta
Dear CASP10 Participants, On the last day of the Meeting we will have our regular Predictors get-together. In advance, I would like to ask you to send in any comments regarding the CASP process in...

Release of CASP10 results
Dear CASP10 Predictors, We have released results of the CASP10 and CASP ROLL experiments. You can check [new interactive results tables](#)

modENCODE (<http://www.modencode.org/>)



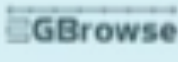
The National Human Genome Research Institute
model organism **ENC**yclopedia Of DNA Elements

[About modENCODE](#) | [Documentation](#) | [Contact Us](#) | [Project Wiki](#) | [Latest News](#)

“The modENCODE Project will try to identify all of the sequence-based functional elements in the *Caenorhabditis elegans* and *Drosophila melanogaster* genomes.”

modMine	Release #32	amazon	Instance	Dataset	FTP
Upload your list of fly genes and view an expression score heatmap.	 Networks  Regions  Maps	The entire modENCODE data set available for analysis in the Amazon compute cloud.	Find, view and download datasets in bulk.	Download released data using the traditional FTP interface.	

Choose an organism below to see GBrowse, Dataset Search links:

[C. elegans](#)[C. brenneri](#)[C. briggsae](#)[C. japonica](#)[C. remanei](#)[D. melanogaster](#)[D. ananassae](#)[D. mojavensis](#)[D. pseudoobscura](#)[D. simulans](#)[D. virilis](#)[D. yakuba](#)

Rfam (<http://rfam.sanger.ac.uk>)



[HOME](#) | [SEARCH](#) | [BROWSE](#) | [FTP](#) | [BIOMART](#) | [BLOG](#) | [HELP](#)



Please note: this site relies heavily on the use of javascript. Without a javascript-enabled browser, this site will not function correctly. Please enable javascript and reload the page, or switch to a different browser.

Rfam 11.0 (August 2012, 2208 families)

The Rfam database is a collection of RNA families, each represented by multiple sequence alignments, consensus secondary structures and covariance models (CMs). [More...](#)

QUICK LINKS

[SEQUENCE SEARCH](#)

[VIEW AN RFAM FAMILY](#)

[VIEW AN RFAM CLAN](#)

[KEYWORD SEARCH](#)

[TAXONOMY SEARCH](#)

JUMP TO

YOU CAN FIND DATA IN RFAM IN VARIOUS WAYS...

Analyze your RNA sequence for Rfam matches

View Rfam family annotation and alignments

view Rfam clan details

Query Rfam by keywords

Fetch families or sequences by NCBI taxonomy

[Go](#) [Example](#)

Enter any type of accession or ID to jump to this page for a Rfam family, sequence or genome

Or view the help pages for more information

Citing Rfam

If you find Rfam useful, please consider citing the references that describe this work:

Rfam 11.0: 10 years of RNA families, R. S.W. Burge, J. Daub, R. Eberhardt, J. Tate, L. Barquist, E.P. Nawrocki, S.R. Eddy, P.P. Gardner, A. Bateman.

Nucleic Acids Research (2012) doi: 10.1093/nar/gks1005

Mirrors

The following are official Rfam mirror sites:

WTST, UK

JFRC, USA

Bioconductor (<http://bioconductor.org/>)

The screenshot shows the Bioconductor website homepage. At the top, there is a teal navigation bar with the Bioconductor logo on the left, which includes the text "BIOCONDUCTOR" and "OPEN SOURCE SOFTWARE FOR BIOLOGICALS". To the right of the logo is a search bar with the placeholder text "Search:". Further right in the navigation bar are links for "Home", "Install", "Help", "Developers", and "About".

Below the navigation bar, the main content area is divided into several sections. On the left, there is a section titled "About Bioconductor" with a sub-header "About Bioconductor". The text below this header describes Bioconductor as a tool for the analysis and comprehension of high-throughput genomic data, mentioning its use of the R statistical programming language and its open-source nature. It also notes that it has two releases each year, 2009 software packages, and an active user community. A link to the "Bioconductor FAQ" is provided.

To the right of the "About" section is a section titled "Use Bioconductor for...". This section contains three sub-sections, each with a green arrow icon: "Microarrays" (describing import, quality assessment, normalization, differential expression, clustering, and other workflows), "High Throughput Assays" (describing import, transform, edit, analyze, and visualize flow cytometry, mass spec, HT-PCR, cell-based, and other assays), and "Recent Courses" (exploring material from recent courses, including BioC2013, BioC2012, BioC2011, Intermediate R / Bioconductor for High Throughput Sequence Analysis).

Below the "Use Bioconductor for..." section, there is a "Mailing Lists" section with a "Subscribe" button and a "Search / post" button. To the right of the "Mailing Lists" section is an "Events" section with a "Subscribe" button. The "Events" section lists several upcoming events: "BioC2014" (10 July - 01 August 2014 - Boston, USA), "Bioconductor European Developers' Workshop" (02 - 10 December 2013 - Cambridge, UK), "Next Generation Data Analysis" (12 - 16 December 2013 - Riverside, CA, USA), and "Summer Bioinformatics Course".

BioPerl (<http://bioperl.org/>)



Main Links

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- Recent changes
- Random page

documentation

- Quick start
- FAQ
- HOWTO's
- API Docs
- Scriptbook
- Tutorials
- Deobfuscator
- Browse Modules

community

- News
- Mailing lists
- Supporting BioPerl
- bioPerl Media
- Hot topics
- About this site
- News
- Mailing lists
- Supporting bioPerl
- BioPerl Media
- Hot Topics

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Main Page

Welcome to BioPerl, a community effort to produce Perl code which is useful in biology.
For more background on the BioPerl project please see the History of BioPerl.
BioPerl is distributed under the [Perl Artistic License](#). For more information, see [Joining BioPerl](#).

Installation	Documentation	Support
- Linux - Windows - Mac OS X - Ubuntu Server - FreeBSD - Fedora	- API Docs and BioPerl docs/P - HOWTO - Scriptbook - The (in)famous Deobfuscator.g	- FAQ - IRC : #bioperl on Webchat.us - Mailing lists - Search mail list archives.g - BioPerl Media options
Developers	How Do I...?	BioPerl-related Distributions
- Using Git - Advanced BioPerl - The SeqIO Modules - Features and Annotations in BioPerl - Writing BioPerl Tests - BioPerl Best Practices - Browse the Core Modules - Bugs	...install BioPerl? ...find a nice, readable BioPerl overview? ...get help? ...learn Perl? ...read a sequence file? ...parse a BLAST/FASTA search? ...get genomic sequences and coordinates? ...query GenBank sequence/annotations? ...contribute code to BioPerl? ...submit a bug report? ...see open BioPerl bugs? 🐛	- Core - BioSQL adapters (BioPerl-DB) - BioPerl-Run wrappers - Network analysis (BioPerl-Network) - Bio::Graphics.g - BioPerl-Pedigree - Gbrowse

- BioPerl v.1.6.910 released
- Travis CI for Testing
- BioPerl-DB, BioPerl-Run, BioPerl-Network 1.0.0 released
- BioPerl 1.0.0 released
- ORF and Google Summer of Code 2011
- Introduction of OpenID logins for ORF wikis
- ORF Hosting server now available
- BioPerl has moved to GitHub
- ORF Google Summer of Code Accepted Students
- ORF in Google Summer of Code

See also our news page, and twitter.g.

BioPython (<http://biopython.org>)



Navigation

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Biopython

Introduction

Biopython is a set of freely available tools for biological computation written in Python by an international team of developers. It is a distributed collaborative effort to develop Python libraries and applications which address the needs of current and future work in bioinformatics. The source code is made available under the Biopython License, which is extremely liberal and compatible with almost every license in the world. We work along with the Open Bioinformatics Foundation, who generously host our website, bug tracker, and mailing lists.

This wiki will help you download and install Biopython, and start using the libraries and tools.

Get Started

- Download Biopython
- Installation help (PDF)

Get help

- Tutorial (PDF)
- Documentation on this wiki
- Cookbook (working examples)
- Discuss and ask questions

Contribute

- What's being worked on
- Developing on Github
- Google Summer of Code
- Report bugs (older issues)

- Biopython 1.63 beta released
- Biopython 1.62 released
- Biopython 1.51 released
- Travis-CI for Testing
- Biopython 1.40 released
- Students selected for OSoC
- Cross-links in GenomDiagram
- Biopython 1.59 released
- Chromosome Diagrams in Biopython
- Biopython 1.58 released

See also our news page, and twitter.

The latest release is Biopython 1.62, released on 26 August 2013. A beta version of 1.63 is also available.

This page was last modified on 12 November 2013, at 19:46.
This page has been accessed 1,283,180 times.
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THANK YOU